

Decomposing a graph into linearly many cycles and edges

Claude Opus 5.5

An AI-generated exposition of the proof claimed by Ryan Coffey [5]

1 October 2026

Status. This write-up is entirely AI-generated. It expounds a proof claimed by Ryan Coffey [5]. An earlier condensed draft was produced by another AI system. The present text, its figures and its checks are by Claude Opus 5.5, with independent AI reviews. At the time of posting (1 October 2026) no human has read or understood this proof, and it has not been refereed or formally verified. It is meant as a stopgap: an intermediate artifact between Coffey’s proof claim and a final proof that humans have read, understood and digested. It should not be cited as an established result.

Abstract

Erdős and Gallai conjectured in the 1960s that the edges of every graph on n vertices can be partitioned into $O(n)$ cycles and single edges. Bucić and Montgomery proved the bound $O(n \log^* n)$. Their argument works in rounds: each round finds well-connected subgraphs, called expanders, turns the edges inside them into few cycles, and passes the remaining edges, which form a much sparser graph, to the next round. There are about $\log^* n$ rounds, and each costs $O(n)$. This paper gives a complete account, in reorganised and simplified form, of a proof of the full conjecture claimed by Coffey. The key observation is that an expander found in one round is far denser than everything that remains two rounds later. So it can lend a few of its edges to every later round, where they serve as connections. With this help the total cost over all rounds is $O(n)$, instead of $O(n)$ per round. The few edges that still cannot be closed into cycles are compressed into auxiliary graphs with at most $n/4$ vertices in total, and these are decomposed by induction.

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1 Introduction

All graphs in this paper are finite and simple. For a graph G let $f(G)$ be the least number of cycles and single edges whose edge sets partition $E(G)$. In the 1960s Erdős and Gallai conjectured that $f(G) = O(n)$ for every graph G on n vertices; see [7] and Problem #184 of [3]. The bound would be best possible up to the constant: cycles contribute even degrees, so in a graph in which every vertex has odd degree, every vertex is an end of a single edge of the decomposition, and at least $n/2$ single edges are needed.

Repeatedly removing a longest cycle gives $f(G) = O(n \log n)$ (Lemma 2.2). Lovász [17] proved that every graph on n vertices decomposes into at most $n/2$ paths and cycles, and Pyber [19] proved that the edges of every such graph can be *covered* by $n - 1$ cycles and edges. For the decomposition problem, Conlon, Fox and Sudakov [6] proved $f(G) = O(n \log \log n)$, and Bucić and Montgomery [4] proved $f(G) = O(n \log^* n)$, where $\log^*$ is the iterated logarithm. This paper gives a complete exposition of a claimed proof of the conjecture; see the note on its status below.

Theorem 1.1. *There is an absolute constant C such that $f(G) \leq Cn$ for every graph G on n vertices.*

Provenance. The argument below is a reorganisation and simplification of a proof claimed by Coffey [5], which in turn builds on the method of Bucić and Montgomery [4]. Coffey reports that his proof was developed with substantial help from an AI system and that it comes with a formalisation in Lean. That formalisation concerns his original manuscript, not the version given here, which, as stated in the box above, is AI-written and has not yet been read by any human. Section 7 lists the source of every ingredient.

1.1 The strategy

This subsection explains the proof informally. Nothing in it is used later.

Expanders. The basic tool is a weak but robust notion of a well-connected graph. Roughly, a graph on N vertices is an *expander* if every set U of at most $2N/3$ vertices has at least $|U|/\log^2 N$ neighbours outside U , and this stays true even after many edges at U are deleted; Definition 2.4 gives the precise form. Such expansion is weak (“sublinear”), but it has two advantages. First, every graph can be cut into expanders that overlap little, at the price of

deleting a few edges (Section 3.2). Second, expanders are richly connected. They contain long cycles (Lemma 3.1). And given many pairs of vertices, an expander contains edge-disjoint paths joining all the pairs, even if the interior of every path must lie in a prescribed sparse random set of vertices (Lemma 4.3).

This second property is how all our cycles are made. To turn a path P into a cycle, we join its two ends by a *connector*, a path through an expander whose interior lies in a random set that P avoids. The connector then meets P only at its ends, so P and the connector together form a cycle.

Rounds. The method of Bucić and Montgomery [4] runs in rounds. A round starts from a graph $G_r \subseteq G$, on all n vertices of G , with average degree $d = d_r$; write $\lambda = \log d$. First one removes cycles of length at least $d\lambda^2$, one at a time; there are at most $n/(2\lambda^2)$ of them. The graph that remains has no long cycles, so all its expanding subgraphs are small, since expanders contain long cycles. One then splits it into expanders, called *pieces*, of order at most d^2 , deleting a few edges along the way. The deleted edges, together with the edges of a few small parts, form the graph G_{r+1} of the next round, whose average degree is only polylogarithmic in d . After about $\log^* n$ rounds the remaining graph has bounded average degree, and its edges are simply taken as single edges.

In [4] the edges inside a piece X are decomposed into paths, and the paths are closed into cycles by connectors inside X . This costs $O(|X|)$ cycles and edges. Summed over the pieces of a round, the cost is $\Theta(n)$ per round. This is where the factor $\log^* n$ comes from: a vertex can lie in pieces of every round, and it is paid for every time.

Anchors. The key observation, due to Coffey [5], is that earlier pieces are vastly richer than anything that happens two rounds later. Every vertex of a piece of round r has degree more than about λ_r^{50} inside the piece, and the piece stays an expander after deleting up to λ_r^{50} edges per vertex (the exponents are chosen in Section 3.3). A piece of round $\ell \geq r + 2$ has at most $M_\ell \approx d_\ell^2$ vertices, and two rounds of polylogarithmic decay give $M_\ell \leq (\log \lambda_r)^{O(1)}$ (Lemma 3.5). So a piece Y of round r can set aside, for each round $\ell \geq r + 2$, small random slices of its edges, called *reservoirs*. Each reservoir is still an expander, with degrees far larger than anything in round ℓ , and round ℓ borrows from it. The remaining half of the edges of Y are its *own* edges, which Y keeps for itself. One round back is not enough, because a piece of round $r + 1$ may have up to λ_r^{110} vertices.

Fix a round ℓ . A vertex of a piece of round ℓ plays one of three roles (Figure 1.1):

- a *hub* lies in at least two pieces of round ℓ . Hubs are rare, because they lie in the separators of the splitting, and separators are tiny: summed over all rounds, the number of incidences between hubs and pieces is $o(n)$;
- a *port* lies in exactly one piece of round ℓ , and also lies in a piece of some round $r \leq \ell - 2$, which we fix and call its *anchor*;
- a *fresh* vertex lies in exactly one piece of round ℓ and has no anchor. Two rounds after a vertex first appears in a piece, it has an anchor, so every vertex is fresh in at most two rounds. A single round, however, may have $\Theta(n)$ fresh vertices.

The edges of round ℓ are decomposed in four steps: clearing, closing cherries, pairing at fresh vertices, and the quotient. We describe them in this order.

Step 1: clearing. Each piece X is an expander, so it can absorb all edges among any chosen set P of its vertices into $O(|P|)$ cycles and edges, using connectors made of its own edges (Lemma 5.1). We clear the hubs and the fresh vertices of X in this way. Hubs are rare

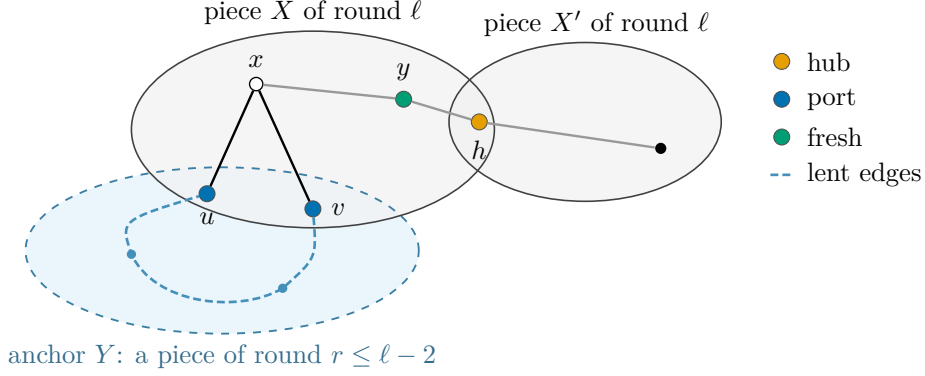


Figure 1.1: Roles of vertices in round ℓ . The hub h lies in two pieces of round ℓ . The ports u, v lie only in X , and both have the old piece Y as their anchor. The edges xu, xv of round ℓ leave Y , and together they form a cherry uxv with both ends in Y (see Step 2 below). A path of edges lent by Y (dashed) closes the cherry into a cycle. The vertex y is fresh.

and every vertex is fresh in at most two rounds, so over all rounds this costs $O(n)$. Afterwards every remaining edge of round ℓ has an end at a port.

Step 2: closing cherries. An edge of round ℓ never has both ends in an earlier piece, because such edges were assigned to that piece when it was formed. So if u is a port with anchor Y , every edge ux of round ℓ leaves Y . Fix Y and a vertex $x \notin Y$, and pair up the edges from x to ports anchored at Y . This gives *cherries* uxv with centre x and both ends in Y . Closing each cherry by its own connector inside Y would cost one cycle for every two edges, which is far too many. Instead, connectors from the reservoirs of Y join the last vertex of one cherry to the first vertex of the next, and so chain K vertex-disjoint cherries into a single cycle (Figure 5.2). With K large, almost all edges at ports turn into few cycles. What remains is at most one unpaired edge from each hub or fresh vertex x to the ports of each anchor, together with some edges between two ports. So the leftover edges are of three kinds: hub–port, fresh–port and port–port.

Step 3: pairing at fresh vertices. A fresh vertex x may still have many leftover edges, one to the ports of each of many anchors. We pair them up at x into paths uxv between two ports, and pay for one edge if their number is odd. This costs at most one edge per fresh vertex, so $O(n)$ over all rounds. These paths, and the leftover edges between two ports, are the *links*. Fresh vertices have to disappear at this point, because there may be $\Theta(n)$ of them, far too many to become vertices of the small auxiliary graphs of the last step.

Step 4: the quotient. What remains are *hub edges* hu from a hub to a port, and links between two ports. We extend every hub edge hu by a private edge uw , lent by the anchor of u , to a vertex w of a sparse random *pool*, and regard the path $h u w$ as a single edge hw of an auxiliary graph Q_ℓ whose vertices are hubs and pool vertices. Links are treated in the same way, with a pool vertex at each end. A cycle of Q_ℓ lifts to a cycle of G , and a single edge of Q_ℓ lifts to at most two edges of G (Figure 5.3). The point is that Q_ℓ may have many edges but has few vertices: hubs are rare and pools are sparse, so the graphs Q_ℓ have at most $n/4$ vertices in total. By induction, a graph on q vertices decomposes into at most Cq cycles and edges, however many edges it has, and lifting at most doubles the number of objects.

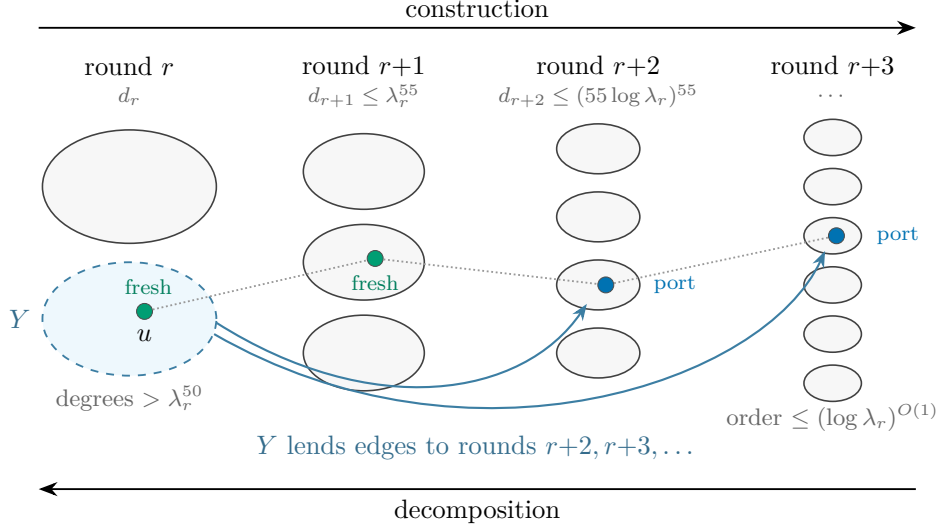


Figure 1.2: The rounds. Pieces get smaller from round to round, because the average degree drops from d to at most $(\log d)^{55}$. A vertex u is fresh (green) in the first two rounds in which it lies in a piece. After that it is a port (blue) whose anchor Y comes from at least two rounds earlier. Every piece lends small random parts of its edges to all rounds at least two steps later. The rounds are constructed from left to right and decomposed from right to left, so every lender is decomposed after all of its borrowers.

The budget. Together with the long cycles and the edges of the final sparse graph, these steps decompose $E(G)$. The objects are:

- the edges of the final sparse graph, fewer than $Dn/2$, where D is the average degree at which the rounds stop;
- $O(n)$ objects for clearing and pairing at fresh vertices;
- $o(n)$ objects for the long cycles, for clearing hubs, for closing cherries, and for a few unlucky ports for which the random choices fail and whose edges are cleared or paid for instead;
- at most $2C \cdot n/4 = Cn/2$ objects for the lifted decompositions of the quotients.

The terms of the third kind are $o(n)$ in total, not just $o(n)$ per round, because the parameters grow like a tower towards earlier rounds, so quantities that decay with the parameters of a round are summable over all rounds (Lemma 3.5). If the first three kinds cost at most C_0n , then $f(G) \leq C_0n + Cn/2 \leq Cn$ for $C = 2C_0$. This is Theorem 1.2 below.

Building forwards, decomposing backwards. All rounds are constructed first. The decomposition then goes through the rounds in reverse order (Figure 1.2). So when a piece Y is finally decomposed, every round that borrowed from Y is finished. Only the edges that were actually borrowed are removed from Y . All other edges of Y , including unused reservoir edges, are decomposed together with its own edges.

The argument thus reduces the problem to graphs with a quarter as many vertices. The precise statement is as follows.

Theorem 1.2 (Reduction). *There is an absolute constant C_0 such that for every graph G on n vertices there are graphs $Q_1, \dots, Q_k$ with*

$$\sum_i |V(Q_i)| \leq n/4 \quad \text{and} \quad f(G) \leq C_0n + 2 \sum_i f(Q_i).$$

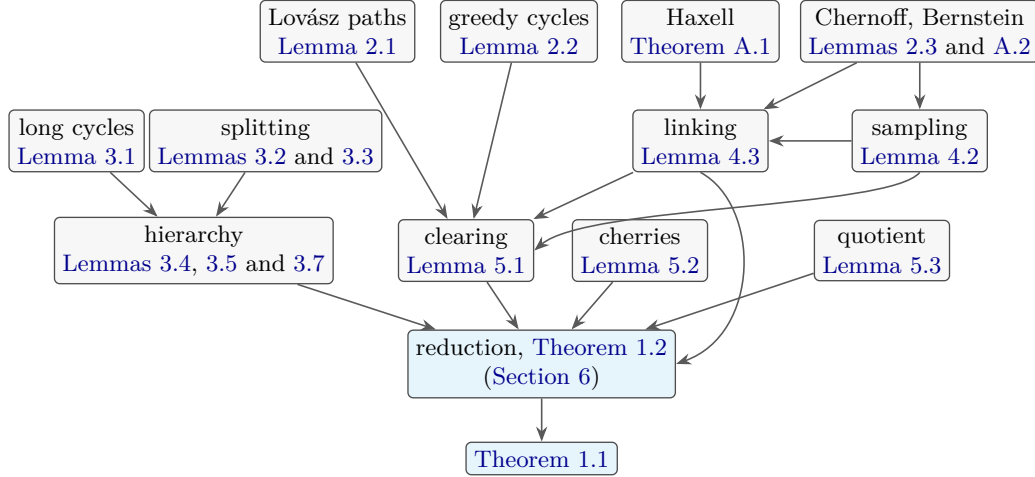


Figure 1.3: How the results depend on each other. The three decomposition lemmas of Section 5 are independent of the hierarchy; they meet only in the assembly of Section 6, which also uses the linking lemma (and with it the sampling lemma) to equip the pieces with reservoirs. The clearing lemma also uses the sampling lemma directly.

Proof of Theorem 1.1 from Theorem 1.2. We show $f(G) \leq 2C_0n$ by induction on n . The case $n = 0$ is trivial. For $n \geq 1$ each Q_i has at most $n/4 < n$ vertices, so by induction $f(G) \leq C_0n + 2 \sum_i 2C_0|V(Q_i)| \leq C_0n + C_0n$. $\square$

1.2 Organisation

Figure 1.3 shows how the results depend on each other. Section 2 fixes notation, recalls three classical tools and defines expanders. Section 3 builds the rounds and proves their structural properties. Section 4 states the linking lemma, which is behind every connection we make through an expander; its proof, an adaptation of the method of [4], is in Appendix A, together with the two tools only it needs (Haxell’s theorem and a Bernstein inequality). Section 5 proves three self-contained decomposition lemmas: clearing a set of vertices, closing cherries, and building the quotient. Section 6 puts everything together and proves Theorem 1.2; it ends with a list of what the proof needs from the exponents. Section 7 lists the sources.

A reader who wants the main line of the argument can read Section 1.1, the statements in Sections 3 and 4, Section 5 and Section 6, and skip the appendix and most proofs of Section 3. We recommend reading the short proof of Lemma 3.7, which shows why everything charged to the rounds adds up to $o(n)$.

2 Preliminaries

2.1 Notation

For a graph H we write $|H|$ for its number of vertices and $e(H)$ for its number of edges. For $U \subseteq V(H)$, $H[U]$ is the induced subgraph and

$$N_H(U) = \{v \in V(H) \setminus U : v \text{ has a neighbour in } U\}$$

is the *external* neighbourhood of U . For disjoint $A, B \subseteq V(H)$, $E_H(A, B)$ is the set of edges with one end in A and the other in B . For a set F of edges, $H - F$ is obtained by deleting F and keeping all vertices, and $\deg_F(v)$ is the number of edges of F at v . For a set E of edges and a set P of vertices, $E[P]$ is the set of edges of E with both ends in P . All logarithms are to base two; $\ln$ is the natural logarithm.

An *object* is a cycle or a single edge, identified with its edge set. A *decomposition* of a set F of edges is a partition of F into objects, and $f(F)$ is the least number of objects in a decomposition of F ; for a graph H , $f(H) = f(E(H))$. Clearly $f(F) \leq |F|$, and $f(F \cup F') \leq f(F) + f(F')$ for disjoint F, F' . To *pay* for an edge means to use it as a single-edge object.

A path is *through* a set V if all its interior vertices lie in V . Its ends are unrestricted, and a single edge is a path through every set.

2.2 Three classical tools

Lemma 2.1 (Paths; Lovász [17], see [4, Corollary 22]). *The edges of every graph can be decomposed into paths so that every vertex is an end of at most two of the paths. In particular, if every edge of a graph has both ends in a set W , then its edges decompose into at most $|W|$ paths.*

The second statement follows from the first because every path has two distinct ends.

Lemma 2.2 (Greedy cycle removal). *Every graph H on $h \geq 1$ vertices satisfies $f(H) \leq h(\log h + 1)$.*

Proof. If a graph on h vertices has $m > h$ edges, then it contains a cycle of length greater than m/h . Indeed, delete vertices of degree less than m/h one at a time. Each deletion removes fewer than m/h edges and at most h vertices are deleted, so fewer than m edges are removed, and the process stops at a nonempty subgraph of minimum degree at least $m/h > 1$. All neighbours of the first vertex v_0 of a longest path $v_0 v_1 \cdots v_k$ in this subgraph lie on the path. If v_i is the last of them, then $i \geq m/h$, and $v_0 v_1 \cdots v_i v_0$ is a cycle of length $i + 1 > m/h$.

Now remove such cycles from H while it has more than h edges. While the number m of remaining edges lies in $(2^i h, 2^{i+1} h]$, every removed cycle has more than 2^i edges, so at most h cycles are removed. As $m \leq h^2/2$, only the ranges with $2^i < h/2$ occur, and there are at most $\log h$ of them. So at most $h \log h$ cycles are removed, and at most h edges remain, which we pay for. $\square$

This is the classical bound $f(G) = O(n \log n)$, observed already by Erdős and Gallai; see [6, 4].

Lemma 2.3 (Chernoff bounds; see [12, Theorems 2.1 and 2.8]). *Let X be a sum of independent indicator variables and $0 < \delta \leq 1$. If $\mathbb{E}X \geq \mu$ then $\mathbb{P}(X \leq (1 - \delta)\mu) \leq e^{-\delta^2 \mu/2}$, and if $\mathbb{E}X \leq \mu$ then $\mathbb{P}(X \geq (1 + \delta)\mu) \leq e^{-\delta^2 \mu/3}$.*

2.3 Robust sublinear expanders

In an expander in our sense, every set U of at most two thirds of the vertices has at least a $1/\log^2$ fraction of $|U|$ as neighbours outside U (“sublinear” expansion), even after up to s edges per vertex of U are deleted (“robust” expansion). Throughout the paper we fix

$$\varepsilon = 1/32.$$

Definition 2.4 (Expanders). Let $s \geq 0$. A graph H on $N \geq 2$ vertices is an *s -expander* if

$$|N_{H-F}(U)| \geq \frac{\varepsilon|U|}{\log^2 N} \quad \text{whenever } 1 \leq |U| \leq \frac{2}{3}N \text{ and } F \subseteq E(H), |F| \leq s|U|.$$

Sublinear expanders were introduced by Komlós and Szemerédi [13, 14]. The robust form, in which up to $s|U|$ edges may be deleted, was introduced independently by Haslegrave, Kim and Liu [10] and by Sudakov and Tomon [20]. In the notation of [4, Definition 11], our s -expanders are the (ε, s) -expanders with $\varepsilon = 2^{-5}$. See [16] for a survey. We use two simple consequences of the definition.

- An s -expander has minimum degree greater than s . Indeed, for a vertex v , taking $U = \{v\}$ and F the set of edges at v gives $N_{H-F}(U) = \emptyset$, so $|F| > s$. In particular $N > s + 1$.
- An s -expander is an s' -expander for every $s' \leq s$.

3 Expanders and the hierarchy of rounds

This section builds the rounds. Sections 3.1 and 3.2 prove the two facts about expanders that the rounds rest on: expanders contain long cycles, and every graph splits into expanders that overlap little. Section 3.3 defines the rounds, and Section 3.4 defines the roles of vertices and proves that the quantities charged to them are summable.

3.1 Long cycles

Bucić and Montgomery use a depth-first search to find a cycle of length $\Omega(N/\log^4 N)$ in a 0-expander on N vertices [4, Lemma 25]. A slightly different use of depth-first search, a now standard tool for long paths and cycles [15], gives a better bound with less work.

Lemma 3.1 (Long cycles in expanders). *Every 0-expander H on $N \geq 2^{20}$ vertices contains a cycle of length at least $N/(2^8 \log^2 N)$.*

Proof. H is connected: otherwise a smallest component K has $|K| \leq N/2$ and $N_H(K) = \emptyset$, which contradicts expansion. Let T be a depth-first search tree of H with root o . Every edge of H joins a vertex to one of its ancestors in T . For a vertex y let T_y be the set of its descendants, including y .

Put $k = \lfloor N/2 \rfloor$, and let v be a vertex of maximum depth with $|T_v| > k$. The sets T_c , for the children c of v , partition $T_v \setminus \{v\}$. Each has size at most k , and together they have size at least k . So we can choose a union U of some of them with $k/2 \leq |U| \leq k$: take one T_c of size at least $k/2$ if there is one, and otherwise add sets T_c one at a time until the size first reaches $k/2$.

Since U is a union of whole subtrees hanging from v , every vertex outside U with a neighbour in U is an ancestor of that neighbour, and hence lies on the path from o to v in T (Figure 3.1). Note that v itself lies in $N_H(U)$. As $|U| \leq N/2$, expansion gives

$$q := |N_H(U)| \geq \frac{\varepsilon k}{2 \log^2 N} \geq \frac{N}{2^8 \log^2 N} \geq 2.$$

Let x be the vertex of $N_H(U)$ closest to o , and let $u \in U$ be a neighbour of x . As $q \geq 2$, x is a proper ancestor of v . So the path from x to u in T has length at least two, and it passes through all q vertices of $N_H(U)$. Together with the edge ux it forms a cycle of length at least $q + 1$. $\square$

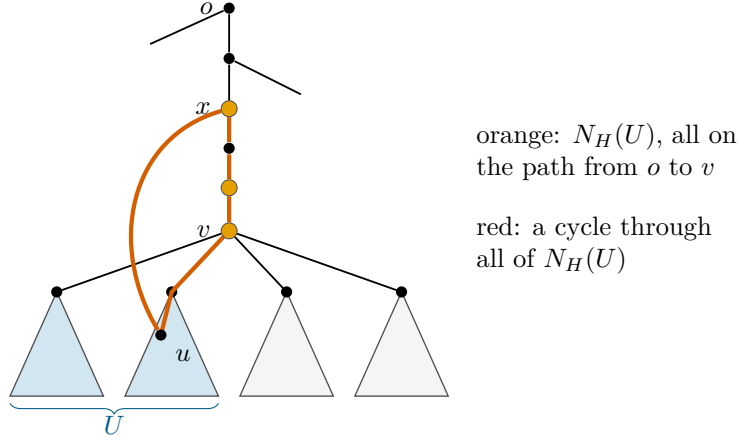


Figure 3.1: Proof of Lemma 3.1. In a depth-first search tree there are no edges between different subtrees of v . So the external neighbourhood of the shaded union U of subtrees lies on the path from the root o to v , and the highest neighbour x closes a cycle through all of it.

3.2 Splitting into expanders

A graph that is not an s -expander has a set U with a small external neighbourhood once at most $s|U|$ edges are deleted. Such a set splits the graph into two smaller graphs that overlap only in a small separator. Repeating this gives a decomposition into expanders. This is the method of Bucić and Montgomery [4, Lemma 14]. Coffey [5] added a rule that moves vertices with many deleted edges into the separator. This makes the deleted edges a thin cut, which is property (iii) below. Later it bounds how many ports of one anchor a single vertex can see (Lemma 3.7(c)).

Lemma 3.2 (A splitting step). *Let $s \geq 0$ and $\tau > 0$, and let H be a graph on $N \geq 2$ vertices which is not an s -expander, with $\tau \geq 256s \log^2 N$. Then there is a partition $V(H) = A \sqcup B \sqcup C$ with $A \neq \emptyset$ such that*

- (i) $|B| \leq 2\varepsilon|A|/\log^2 N$ and $|A| + |B| < 3N/4$;
- (ii) the set $F = E_H(A, C)$ of deleted edges satisfies $|F| \leq 2s|A|$, and $F = \emptyset$ if $s = 0$;
- (iii) every vertex of A has fewer than τ neighbours in C , and every vertex of C has fewer than τ neighbours in A .

The two children of H are

$$H_1 = H[A \cup B] \quad \text{and} \quad H_2 = H[B \cup C] - E(H[B]).$$

They meet in the separator B , both have fewer than N vertices, and $E(H)$ is the disjoint union of $E(H_1)$, $E(H_2)$ and F (Figure 3.2).

Proof. Since H is not an s -expander, there are U and F' with $1 \leq |U| \leq 2N/3$, $|F'| \leq s|U|$ and $|N_{H-F'}(U)| < \varepsilon|U|/\log^2 N$. Let $S = N_{H-F'}(U)$ and $C' = V(H) \setminus (U \cup S)$. All edges between U and C' lie in F' , so $F_0 = E_H(U, C')$ has at most $s|U|$ edges. Let U^+ be the set of vertices $u \in U$ with $\deg_{F_0}(u) \geq \tau$, and let $A = U \setminus U^+$. Let C^+ be the set of vertices of C' with at least τ neighbours in A . Put

$$B = S \cup U^+ \cup C^+, \quad C = C' \setminus C^+.$$

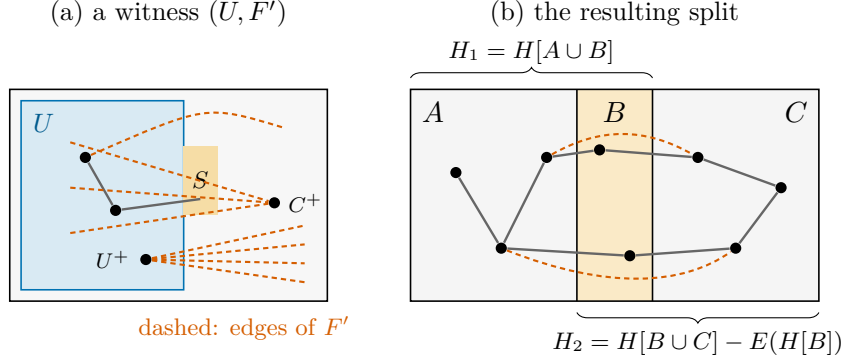


Figure 3.2: A splitting step (Lemma 3.2). (a) After deleting the dashed edges F' , the set U has only the small boundary S . Some vertices would be left with many deleted edges: U^+ inside U , and C^+ outside it. (b) These vertices are moved into the separator $B = S \cup U^+ \cup C^+$. The children H_1 and H_2 overlap only in B , and edges inside B go to H_1 . The deleted edges are now only the few edges between A and C , and they form a thin cut: no vertex has τ or more of them.

The edges of F_0 at vertices of U^+ and the edges between C^+ and A are disjoint subsets of F_0 . Hence $\tau(|U^+| + |C^+|) \leq s|U|$, and so $|U^+| + |C^+| \leq |U|/(256 \log^2 N) = \varepsilon|U|/(8 \log^2 N)$. Thus $|A| \geq \frac{255}{256}|U| > 0$ and

$$|B| < \frac{9\varepsilon|U|}{8 \log^2 N} \leq \frac{2\varepsilon|A|}{\log^2 N}, \quad |A| + |B| \leq |U| + |B| \leq \frac{2}{3}N(1 + \frac{9}{256}) < \frac{3}{4}N.$$

Every edge between $A \subseteq U$ and $C \subseteq C'$ lies in F_0 , so $|F| \leq s|U| \leq 2s|A|$. If $s = 0$ then $F_0 = \emptyset$. Finally, a vertex of A has $\deg_{F_0} < \tau$, and a vertex of C has fewer than τ neighbours in A by the definition of C^+ . $\square$

A *splitting tree* for a graph H_0 is a rooted binary tree whose nodes are graphs and whose root is H_0 . Each node H that is not a leaf is split by Lemma 3.2, for some parameters s_H, τ_H such that H is not an s_H -expander and $\tau_H \geq 256s_H \log^2 |H|$. Its two children are the two children of the split. The edge set of H_0 is the disjoint union of the edge sets of the leaves and the sets of deleted edges at all splits.

Every vertex of a node lies in some leaf below it, because the children of a node cover its vertex set. A vertex lies in two or more leaves if and only if it lies in the separator of some split. Indeed, a vertex of a separator lies in both children, and a vertex lying in two leaves lies in both children of their lowest common ancestor. Such vertices are *shared*, and the others are *private*. For a leaf Y we write $\text{sh}(Y)$ for its set of shared vertices. A private vertex of Y lies in exactly the nodes on the path from the root to Y .

Lemma 3.3 (Splitting trees). *Let $\mathcal{T}$ be a splitting tree for a graph H_0 on n_0 vertices.*

- (a) *The leaves of $\mathcal{T}$ have total order less than $4n_0/3$.*
- (b) *For every $\gamma \geq 2$, $\sum_Y |\text{sh}(Y)| < n_0/\log \gamma$, where the sum is over the leaves Y with $|Y| \geq \gamma$.*
- (c) *If $s_H \leq s$ for every split, then at most $7sn_0 \log n_0$ edges are deleted in total.*
- (d) *Suppose that $\tau_H \leq \tau$ at every split that deletes at least one edge. Then for every leaf Y and every vertex h , fewer than τ deleted edges join h to private vertices of Y , and none do if $h \in Y$.*

The idea is to charge the separator B of each split, and its deleted edges, to the vertices of the side A . A vertex can be charged repeatedly, but each time by a node at most $3/4$ as large as the previous one, so the charges to it form a convergent series. Since a shared vertex lies in several leaves, we charge pairs (leaf, vertex), called *copies*, rather than vertices.

Proof. For every split of a node H , with parts A, B, C , and every $a \in A$, choose a leaf below the first child that contains a , and call the pair (leaf, a) a *copy*. Consider a fixed copy (Y, v) , and let $H^{(1)} \supsetneq H^{(2)} \supsetneq \dots$ be the nodes whose splits chose it. Each $H^{(i+1)}$ lies below the first child of $H^{(i)}$, so $|H^{(i+1)}| < \frac{3}{4}|H^{(i)}|$. Consequently, if all these nodes have order at least $\gamma \geq 2$, then the j -th smallest has order at least $\gamma(4/3)^{j-1}$.

(a),(b) Let Σ_γ be the sum of $|B|$ over the splits of nodes of order at least γ . Spread the bound $|B| \leq \sum_{a \in A} 2\varepsilon / \log^2 |H|$ of [Lemma 3.2](#) over the copies chosen at the split. A copy then receives at most

$$2\varepsilon \sum_{j \geq 0} \frac{1}{(\log \gamma + j \log \frac{4}{3})^2} \leq 2\varepsilon \left(\frac{1}{\log^2 \gamma} + \frac{1}{\log \frac{4}{3} \cdot \log \gamma} \right) < \frac{7\varepsilon}{\log \gamma}$$

from splits of order at least γ . Replacing a node by its two children increases the total order by $|B|$, so the total order of the leaves is $S = n_0 + \Sigma_2$. As there are at most S copies, $\Sigma_\gamma < 7\varepsilon S / \log \gamma$. For $\gamma = 2$ this gives $S < n_0 + 7\varepsilon S$, so $S < n_0 / (1 - 7\varepsilon) < 4n_0/3$, which proves (a). Hence $\Sigma_\gamma < \frac{7}{24}n_0 / \log \gamma$.

For (b), let c_v be the number of leaves of order at least γ in which v is shared, and k_v the number of splits of nodes of order at least γ with v in the separator. We claim $c_v \leq 2k_v$. Suppose $c_v \geq 1$. The leaves counted by c_v , together with their ancestors, form a subtree whose nodes all have order at least γ and contain v . If $c_v \geq 2$, this subtree has at least $c_v - 1$ nodes with two children in it, and v lies in the separator of each of them. If $c_v = 1$, then v also lies in another leaf, and v lies in the separator of the lowest common ancestor of the two leaves, which has order at least γ . Thus $\sum_Y |\text{sh}(Y)| = \sum_v c_v \leq 2\Sigma_\gamma < n_0 / \log \gamma$.

(c) Spread the bound $|F| \leq 2s|A|$ over the copies chosen at the split, $2s$ each. The orders of the nodes charging a fixed copy decrease by a factor $3/4$ and lie between 2 and n_0 , so there are at most $1 + \log_{4/3}(n_0/2) \leq 2.41 \log n_0$ of them. Hence at most $2s \cdot \frac{4}{3}n_0 \cdot 2.41 \log n_0 \leq 7sn_0 \log n_0$ edges are deleted.

(d) Let hy be a deleted edge with y private to Y , deleted at the split of a node H . Since y is private, H is an ancestor of Y , and y lies in the part (A or C) of the child towards Y . So h lies in the other part and not in the child towards Y , and therefore $h \notin Y$. Moreover, H is the last node on the path from the root to Y that contains h . So all the deleted edges in question are deleted at the same split. Since h and the private vertices of Y lie in opposite parts of this split, there are fewer than τ of them by [Lemma 3.2\(iii\)](#). $\square$

3.3 The rounds

We fix a large absolute constant D . Every statement from now on holds if D is larger than a suitable absolute constant, and we assume this tacitly. We write $o_D(n)$ for any quantity bounded by $\eta(D)n$, where η is a function that depends only on D and tends to 0 as $D \rightarrow \infty$.

For a real number $d \geq D$ we use the parameters

$$\lambda = \log d, \quad M = \lceil d^2 \rceil, \quad s = \lceil \lambda^{50} \rceil, \quad \tau = \lceil \lambda^{53} \rceil, \quad m = \lceil \lambda^{54} \rceil,$$

with the meanings shown in [Table 3.1](#). The exponents are not optimised; [Section 6.5](#), at the end of the proof, lists what the proof needs from them.

symbol	value	role
d_r	$2e(G_r)/n$	average degree at the start of round r
λ_r, μ_r	$\log d_r, \log \lambda_r$	logarithms of the degree
M_r	$\lceil d_r^2 \rceil$	maximum order of a piece of round r
s_r	$\lceil \lambda_r^{50} \rceil$	pieces of round r are s_r -expanders
τ_r	$\lceil \lambda_r^{53} \rceil$	thin-cut threshold for splits in round r
m_r	$\lceil \lambda_r^{54} \rceil$	minimum order of a piece of round r

Table 3.1: The parameters of round r .

Construction. Let $G_1 = G$. For $r = 1, 2, \dots$ let $d_r = 2e(G_r)/n$. If $d_r < D$ we stop and put $R = r - 1$. Otherwise we carry out round r , with the parameters of Table 3.1:

- (R1) *Long cycles.* Repeatedly remove from G_r a cycle of length at least $d_r \lambda_r^2$, as long as there is one. Let $\mathcal{C}_r$ be the set of removed cycles and G'_r the remaining graph.
- (R2) *Splitting.* Build a splitting tree $\mathcal{T}_r$ for G'_r in two stages. In the first stage, split every node with at least two vertices that is not a 0-expander, with $(s_H, \tau_H) = (0, \tau_r)$. The leaves of the first stage are 0-expanders or single vertices. In the second stage, split every node of order at least m_r that is not an s_r -expander, with $(s_H, \tau_H) = (s_r, \tau_r)$. The leaves of order at least m_r are s_r -expanders and are called the *pieces* of round r . For a piece Y (a set of vertices) we write H_Y for the leaf itself, an s_r -expander with vertex set Y .
- (R3) *Assignment.* Assign every edge of every H_Y to Y . Then assign every other edge of G'_r with both ends in some piece to one such piece. Let $E_Y \supseteq E(H_Y)$ be the set of edges assigned to Y .
- (R4) *Next round.* G_{r+1} consists of the edges of G'_r that were not assigned, on the vertex set $V(G)$.

We write $\mathcal{Y}_r$ for the set of pieces of round r , and $\text{sh}_r(Y)$ for the shared vertices of $Y \in \mathcal{Y}_r$ in $\mathcal{T}_r$.

Each step has a purpose. Removing long cycles in (R1) forces every expander that remains to be small, at most M_r vertices, because expanders contain long cycles (see (H1) below). The splitting in (R2) has two stages because of the deleted edges: the first stage deletes nothing and cuts the graph into parts of order at most M_r , so the second stage, which does delete edges, works inside small parts and deletes only $O(s_r \log M_r)$ edges per vertex rather than $O(s_r \log n)$; see (H5). The assignment in (R3) makes the rounds *separated*: no later round ever sees an edge with both ends in an earlier piece; see (H6). This is what makes cherries possible.

Lemma 3.4 (The hierarchy). *For every round $r \leq R$:*

- (H1) every piece $Y \in \mathcal{Y}_r$ has $m_r \leq |Y| \leq M_r$;
- (H2) $\sum_{Y \in \mathcal{Y}_r} |Y| < 4n/3$, so $|\mathcal{Y}_r| < 4n/(3m_r)$;
- (H3) $\sum_{Y \in \mathcal{Y}_r} |\text{sh}_r(Y)| < n/\log m_r$;
- (H4) $|\mathcal{C}_r| \leq n/(2\lambda_r^2)$;
- (H5) $d_{r+1} \leq \lambda_r^{55}$;
- (H6) (separation) for every $\ell > r$, no edge of G_ℓ has both ends in a piece of round r ;
- (H7) (thin cuts) for every $Y \in \mathcal{Y}_r$ and every vertex h , fewer than τ_r edges of G_{r+1} join h to vertices of $Y \setminus \text{sh}_r(Y)$.

Moreover, $E(G)$ is the disjoint union of the cycles in $\bigcup_r \mathcal{C}_r$, the sets E_Y over all pieces of all rounds, and $E(G_{R+1})$, where $e(G_{R+1}) < Dn/2$. The graphs H_Y over all pieces of all rounds are pairwise edge-disjoint.

Proof. (H1) Let W be a leaf of the first stage, a 0-expander, and suppose $|W| > M_r$. By Lemma 3.1 and since $x/\log^2 x$ is increasing for $x \geq 8$, W contains a cycle of length at least $M_r/(2^8 \log^2 M_r) \geq d_r^2/(2^8 \cdot 9\lambda_r^2) > d_r \lambda_r^2$. But $W \subseteq G'_r$, which has no such cycle. So the first-stage leaves, and hence all pieces, have order at most M_r .

(H2),(H3) These follow from Lemma 3.3(a) and (b) with $\gamma = m_r$.

(H4) Each removed cycle has at least $d_r \lambda_r^2$ of the $d_r n/2$ edges of G_r .

(H5) Every edge of G_{r+1} is deleted in $\mathcal{T}_r$ or lies in a leaf of order less than m_r . The first stage deletes nothing. Below each first-stage leaf W the second stage is a splitting tree in which $s_H \leq s_r$. So by Lemma 3.3(c) it deletes at most $7s_r|W| \log M_r$ edges. The first-stage leaves have total order less than $4n/3$, so at most $7s_r \cdot \frac{4}{3}n(2\lambda_r + 1) \leq 30\lambda_r^{51}n$ edges are deleted. A leaf K of order less than m_r has fewer than $|K|m_r/2$ edges, and all leaves have total order less than $4n/3$. Hence $d_{r+1} \leq 60\lambda_r^{51} + \frac{4}{3}m_r \leq \lambda_r^{55}$.

(H6) Every edge of G'_r with both ends in a piece of round r was assigned in (R3), and $G_\ell \subseteq G_{r+1}$ for $\ell > r$.

(H7) Let $hu \in E(G_{r+1})$ with $u \in Y \setminus \text{sh}_r(Y)$. If hu lay in a leaf of $\mathcal{T}_r$, that leaf would contain u , so it would be Y , and hu would have been assigned to Y . So hu is a deleted edge, and Lemma 3.3(d) applies with $\tau = \tau_r$.

The final statements follow from the construction, since $d_{R+1} < D$. $\square$

Lemma 3.5 (Scales). *Let $\mu_r = \log \lambda_r$.*

- (a) *For $r < R$: $\lambda_{r+1} \leq \lambda_r/2$, $\mu_{r+1} \leq \mu_r/2$ and $M_{r+1} \leq M_r/2$. Also $\mu_R \geq \log \log D$ and $M_R \geq D^2$.*
- (b) *For $3 \leq \ell \leq R$ and $r \leq \ell - 2$: $M_\ell \leq (55\mu_{\ell-2})^{111} \leq (55\mu_r)^{111}$, and $2^{\ell-r} \leq \lambda_r$.*
- (c) *For fixed $a > 0$ and $k \geq 0$, $\sum_{r=1}^R (R-r+1)^k \mu_r^{-a} = O_{a,k}((\log \log D)^{-a})$. The same holds with μ_r replaced by λ_r , m_r or M_r , which are at least μ_r .*

Proof. (a) By (H5), $\lambda_{r+1} \leq 55 \log \lambda_r = 55\mu_r \leq \lambda_r/2$ and $\mu_{r+1} \leq \log(55\mu_r) \leq \mu_r/2$, because $\mu_r \geq \log \log D$ is large. Also $d_{r+1} \leq \lambda_r^{55} \leq d_r/2$, which gives $M_{r+1} \leq M_r/2$. (b) We have $M_\ell \leq 2d_\ell^2 \leq 2\lambda_{\ell-1}^{110} \leq 2(55\mu_{\ell-2})^{110} \leq (55\mu_{\ell-2})^{111}$. By (a), $\lambda_r \geq 2^{\ell-r}\lambda_\ell \geq 2^{\ell-r}$. (c) By (a), $\mu_r \geq 2^{R-r} \log \log D$, so the sum is at most $(\log \log D)^{-a} \sum_{j \geq 0} (j+1)^k 2^{-aj}$. $\square$

Part (c) is the main place where the number of rounds enters. It is why no factor $\log^* n$ appears: quantities charged once per round, or even once per pair of rounds, are summable because the parameters grow like a tower towards earlier rounds. Part (b) is why anchors are taken two rounds back: M_{r+1} may be as large as λ_r^{110} , but M_{r+2} is only polylogarithmic in λ_r .

3.4 Roles

Definition 3.6 (Hubs, residents, ports and anchors). Fix a round $\ell \leq R$. A *hub* of round ℓ is a vertex that lies in at least two pieces of round ℓ . Let Hub_ℓ be the set of hubs. A vertex that lies in exactly one piece X of round ℓ is a *resident* of X . A resident u is a *port* if some piece of a round $r \leq \ell - 2$ contains u . In that case we fix one such piece and call it the *anchor* $\text{anc}_\ell(u)$; this choice depends only on the hierarchy, not on any of the random choices made later. A resident that is not a port is *fresh*, and Fresh_ℓ is the set of fresh residents.

For a piece Y of round r and a round $\ell \geq r + 2$, the *crowding* of Y in round ℓ is

$$\kappa_{Y,\ell} = \max_{h \in V(G)} |\{u : u \text{ is a port of round } \ell \text{ with } \text{anc}_\ell(u) = Y \text{ and } hu \in E(G_\ell)\}|.$$

Crowding measures how many cherries can share a centre. The number of cycles needed to close the cherries of an anchor grows with its crowding (Lemma 5.2), so we need the crowding to be small on average. Let ν_ℓ be the number of pieces of rounds at most $\ell - 2$, that is, the number of possible anchors in round ℓ .

The next lemma bounds everything that the assembly in Section 6 charges to the roles. Each item pays for one cost:

- (a) fresh vertices are cleared and paired at a cost of $O(1)$ each per round (Steps 1 and 3), so they must be few when summed over all rounds;
- (b) hubs are cleared (Step 1) and become vertices of the quotient (Step 4);
- (c) closing the cherries of an anchor Y costs about $\kappa_{Y,\ell}/2$ cycles (Step 2);
- (d) it also costs about $2M_\ell$ cycles per anchor in use, so we need $\sum_\ell \nu_\ell M_\ell = o(n)$;
- (e) the long cycles are objects of the decomposition.

Lemma 3.7 (Bookkeeping).

- (a) $\sum_\ell |\text{Fresh}_\ell| \leq 2n$.
- (b) $\sum_\ell \sum_{X \in \mathcal{Y}_\ell} |X \cap \text{Hub}_\ell| = o_D(n)$.
- (c) $\kappa_{Y,\ell} < \tau_r + |\text{sh}_r(Y)|$ for $Y \in \mathcal{Y}_r$, and $\sum_\ell \sum_{r \leq \ell-2} \sum_{Y \in \mathcal{Y}_r} \kappa_{Y,\ell} = o_D(n)$.
- (d) $\nu_\ell \leq 3n/m_{\ell-2}$, and $\sum_{\ell=3}^R \nu_\ell M_\ell = o_D(n)$.
- (e) $\sum_r |\mathcal{C}_r| = o_D(n)$.

Proof. (a) Let r_0 be the first round in which a vertex u lies in a piece. In every round $\ell \geq r_0 + 2$ in which u is a resident, the piece of round r_0 containing u makes u a port. So u is fresh in at most the two rounds r_0 and $r_0 + 1$.

(b) A hub lies in two leaves of $\mathcal{T}_\ell$, so $X \cap \text{Hub}_\ell \subseteq \text{sh}_\ell(X)$. By (H3) the inner sum is less than $n/\log m_\ell \leq n/(54\mu_\ell)$. Now sum using Lemma 3.5(c).

(c) Let u be a port of round ℓ with anchor $Y \in \mathcal{Y}_r$ and $hu \in E(G_\ell) \subseteq E(G_{r+1})$. If $u \notin \text{sh}_r(Y)$, there are fewer than τ_r such u for a fixed h , by the thin-cut property (H7). All other such u lie in $\text{sh}_r(Y)$. Summing over $Y \in \mathcal{Y}_r$ with (H2) and (H3), and then over the fewer than $R - r$ rounds ℓ , the total is at most

$$\sum_r (R - r) \left(\frac{4n}{3m_r} \tau_r + \frac{n}{\log m_r} \right) \leq \sum_r (R - r) \left(\frac{2}{\lambda_r} + \frac{1}{54\mu_r} \right) n = o_D(n).$$

(d) By (H2) and Lemma 3.5(a), $\nu_\ell < \sum_{r \leq \ell-2} 4n/(3m_r) \leq 3n/m_{\ell-2}$. By Lemma 3.5(b), $\nu_\ell M_\ell \leq 3n(55\mu_{\ell-2})^{111}/\lambda_{\ell-2}^{54} \leq n/\lambda_{\ell-2}$, and the sum is $o_D(n)$ by Lemma 3.5(c).

(e) This follows from (H4) and Lemma 3.5(c). $\square$

4 Linking through sparse random sets

Every cycle we build through an expander is closed by a path through a sparse random set of vertices. This section states that robust expanders support such paths, for many pairs at once. Recall that a path is through a set V if all its interior vertices lie in V .

Definition 4.1 (Linking). Let H be a graph, $V \subseteq V(H)$ and $t \geq 1$. We say that H *links through V with multiplicity t* if the following holds for every multiset $\mathcal{P}$ of pairs of distinct vertices of H in which every vertex lies in at most t pairs, counted with multiplicity: there are pairwise edge-disjoint paths P_π , $\pi \in \mathcal{P}$, in H , such that P_π joins the two vertices of π and is through V .

This is the notion of path connectivity of [4, Definition 7], without the bound on the lengths of the paths, which we never need. First we note that a random sample of the edges of an expander is again an expander.

Lemma 4.2 (Sampling edges). *Let H be an s -expander on N vertices, and let $0 < p \leq 1$ with $ps \geq 100 \log N$. Let H_p be the random spanning subgraph of H that contains every edge independently with probability p . Then H_p is a $(ps/2)$ -expander with probability at least $1 - N^{-10}$.*

Proof. Suppose H_p is not a $(ps/2)$ -expander. Then there are U with $1 \leq u = |U| \leq 2N/3$ and $F \subseteq E(H_p)$ with $|F| \leq psu/2$ such that $T = N_{H_p-F}(U)$ has fewer than $\varepsilon u / \log^2 N$ vertices. All edges of H_p between U and $V(H) \setminus (U \cup T)$ lie in F . On the other hand, H has more than su edges between U and $V(H) \setminus (U \cup T)$: otherwise deleting them would leave the external neighbourhood of U inside T , which contradicts expansion. So for this pair (U, T) , at most $psu/2$ of more than su given edges survive in H_p . By Lemma 2.3 this has probability at most $e^{-psu/8} \leq N^{-18u}$. There are at most N^{2u} pairs (U, T) with $|U| = u$, since $|T| < u$. Hence the failure probability is at most $\sum_{u \geq 1} N^{-16u} \leq N^{-10}$. $\square$

The following lemma is the engine of the proof. Bucić and Montgomery proved it for random sets of density $1/3$ and a fixed polylogarithmic multiplicity [4, Theorem 16]. We need sparse random sets and arbitrary multiplicity.

Lemma 4.3 (Linking). *There is an absolute constant C_* such that the following holds. Let $t \geq 1$ and $0 < \rho \leq 1/2$, let H be an s -expander on N vertices with $s \geq C_* t \rho^{-3} \log^{18} N$, and let $V \subseteq V(H)$ be random, containing every vertex independently with probability ρ . Then with probability at least $1 - N^{-3}$, H links through V with multiplicity t .*

Linking through V implies linking through every superset of V . Hence the lemma also applies if every vertex is included independently with probability at least ρ : a ρ -random set can be coupled inside such a set.

The proof is in Appendix A. It follows the strategy of [4], with two simplifications: a single bounded-degree certificate replaces their two-case analysis [4, Proposition 13], and Haxell's theorem replaces the hypergraph matching theorem of Aharoni and Haxell [1]. Here is its outline. Write $L = \log N$. A *ball* of radius k around a set U is the set of vertices of V that can be reached from U by a path through V of length at most k .

- (1) *Haxell reduces linking to one path at a time.* By Theorem A.1, applied to the families of short paths joining each pair, it suffices to show the following: for any set of demands and any set S of forbidden edges whose size is linear in the number of demands, some demand can still be joined by a short path through V in $H - S$.
- (2) *Large balls give such a path.* If every set of u vertices has a ball of radius k covering more than half of V , then among any $2u - 1$ disjoint demands one can be joined by a path of length $O(kL)$. This halving argument is [4, Proposition 8].
- (3) *Balls grow.* In an expander, a random p -subset of a set W has about $\alpha|W|$ neighbours outside W , where $\alpha = \varepsilon/L^2$. A maximal bounded-degree bipartite subgraph between W and its complement certifies this, and a Bernstein inequality makes it hold with very

high probability. Exposing V in $k = O(L/\alpha)$ independent batches, the ball around a set grows by a factor $1 + \alpha/16$ per batch until it covers most of the graph.

- (4) *Union bound.* If the starting set A is *well-expanding*, with $|N(A)| \geq \theta|A|$ for a suitable θ , the growth fails with probability at most $N^{-\Omega(|A|)}$. This is small enough for a union bound over all such A . Every set contains a large well-expanding subset [4, Proposition 18], so this suffices.
- (5) *Amplification.* Steps (3) and (4) tolerate only about $|U|/L^{O(1)}$ deleted edges, while step (1) needs tolerance of about $t|U|L^{O(1)}$. Split the edges of H at random into K colours, each again an expander by Lemma 4.2. Any set of deleted edges puts at most a $1/K$ fraction of itself into some colour, and that colour alone provides the large ball.

The requirement $s \geq C_* t \rho^{-3} L^{18}$ collects the losses: the multiplicity t and one factor ρ^{-1} enter through step (1), two further factors ρ^{-1} through steps (3) and (4), where one batch of V has density only $\rho/O(L^3)$, and the powers of L come from the sublinear expansion.

5 Three decomposition lemmas

This section proves the three local moves of the proof. They are stated without reference to the hierarchy. In Section 6 they are applied once per piece (Lemma 5.1) and once per round (Lemmas 5.2 and 5.3). In the last two, Δ denotes a bound on degrees; in the application it is M_ℓ .

5.1 Clearing a set of vertices

Lemma 5.1 (Clearing). *There is an absolute constant N_0 such that the following holds. Let O be an s -expander with vertex set X , where $N = |X| \geq N_0$ and $s \geq \log^{40} N$. Let E be a set of edges with both ends in X , and let $P \subseteq X$. Then there is a set $O' \subseteq E(O)$ with*

$$f(E[P] \cup O') \leq 101|P|.$$

In words: at a cost of $O(1)$ cycles and edges per vertex of P , we can decompose all edges of E with both ends in P , together with some edges of the expander O . In the application O consists of edges of E , and every edge of E that is left over has an end outside P .

The proof uses a *vortex*, a nested sequence of random sets $P = P_0 \supseteq P_1 \supseteq \dots \supseteq P_k$ whose sizes halve. This device comes from iterative absorption [2]. At step j we decompose all remaining edges that touch the layer $\Lambda_j = P_j \setminus P_{j+1}$ into paths (Lemma 2.1). We extend each path at both ends so that it ends in the next level, and close it into a cycle by a connecting path through a random set inside the next level (Figure 5.1). Closing paths through path-connecting random sets is the method of [4]. Combining it with a vortex is due to Coffey [5]. What survives all k steps lives on the tiny set P_k and is decomposed by Lemma 2.2.

Two points need care. A connector must avoid the path it closes, so each path is given a *phase* $c \in \{1, 2\}$: it avoids the random set $V_{j,c}$ through which its connector will run. And the linking lemma can join only pairs in which no vertex appears too often, so the ends of the paths must be spread out. This is why each vertex w extends its paths only into a fixed set of *outlets*, chosen so that no vertex is an outlet of too many others.

Proof. Let $L = \log N$, and assume $P \neq \emptyset$.

Outlets. Let $a = \lceil L^6 \rceil$ and $b = \lceil 64aL^2 \rceil = \lceil 2aL^2/\varepsilon \rceil$. We claim that every vertex w has a set $\text{Out}(w)$ of a neighbours in O , its *outlets*, such that every vertex is an outlet of at most b vertices. Form a bipartite graph. On the left are the ordered pairs (w, y) with $wy \in E(O)$.

On the right, every vertex y has b *load* slots and every vertex w has $\deg_O(w) - a \geq 0$ *spare* slots. Join (w, y) to the load slots of y and to the spare slots of w . A matching that covers the left side gives, for every w , at least a pairs (w, y) matched to load slots, and we let $\text{Out}(w)$ consist of a of these vertices y .

We check Hall's condition [9]. Let Ψ be a set of pairs, let Z be the set of their second coordinates, and let S be the set of w that occur in more than $\deg_O(w) - a$ pairs of Ψ . Then $|\Psi| \leq (\text{spare slots}) + a|S|$, so it suffices that $b|Z| \geq a|S|$. This holds trivially if $S = \emptyset$. Otherwise every $w \in S$ has fewer than a neighbours outside Z . Take $S' \subseteq S$ with $|S|/2 \leq |S'| \leq 2N/3$, and delete the at most $(a-1)|S'| \leq s|S'|$ edges from S' to outside Z . Then the external neighbourhood of S' lies in Z , so $|Z| \geq \varepsilon|S'|/L^2 \geq \varepsilon|S|/(2L^2)$, and $b|Z| \geq a|S|$.

The random vortex. Let $k = \lfloor \log(L/6) \rfloor$, so that $6/L \leq 2^{-k} < 12/L$. Make the following independent random choices.

- Colour every edge of O uniformly with one of $2k+1$ colours: a *reserve* colour and *connector* colours (j, c) , for $0 \leq j < k$ and $c \in \{1, 2\}$. Let O^{res} and $O_{j,c}$ be the colour classes.
- Give every $v \in P$ a level $\text{lev}(v) \in \{0, \dots, k\}$ with $\mathbb{P}(\text{lev}(v) \geq j) = 2^{-j}$, and put $\text{lev}(v) = k$ for $v \notin P$. Let $U_j = \{v : \text{lev}(v) \geq j\}$, $P_j = P \cap U_j$ and $\Lambda_j = P_j \setminus P_{j+1}$.
- Give every vertex a label in $\{0, 1, 2\}$, with probabilities $\frac{1}{2}, \frac{1}{4}, \frac{1}{4}$. For $c \in \{0, 1, 2\}$ let $V_{j,c}$ be the set of vertices of U_{j+1} with label c .

Thus $V_{j,0}$ will receive the reserved edges, and $V_{j,1}, V_{j,2}$ will carry the connectors of the two phases. We claim that with positive probability the following hold:

- (V1) every $O_{j,c}$ with $c \in \{1, 2\}$ links through $V_{j,c}$ with multiplicity $b+2$;
- (V2) for every $w \in P$ and $j < k$, at least four edges of O^{res} join w to $\text{Out}(w) \cap V_{j,0}$;
- (V3) $\sum_{j < k} |P_j| \leq 8|P|$ and $|P_k| \leq 48|P|/L$.

By Lemma 4.2, each $O_{j,c}$ is an $s/(4k+2)$ -expander with probability at least $1 - N^{-10}$. Every vertex lies in $V_{j,c}$ independently with probability at least $2^{-k}/4 \geq 1/L$. So Lemma 4.3, with $t = b+2 = O(L^8)$ and $\rho = 1/L$, gives (V1) with probability $1 - O(kN^{-3})$; it needs $s/(4k+2) \geq C_*(b+2)L^{21}$, which holds as $s \geq L^{40}$. For (V2), the number of such edges is a sum of independent indicators, one for each $y \in \text{Out}(w)$, with mean at least $a \cdot \frac{1}{2k+1} \cdot \frac{6}{L} \cdot \frac{1}{2} \geq L^4$. So by Lemma 2.3 it is less than four with probability at most $e^{-L^4/3}$, and a union bound over the Nk pairs (w, j) costs $o(1)$. For (V3), the means are $\sum_j 2^{-j}|P| < 2|P|$ and $2^{-k}|P| < 12|P|/L$, so by Markov's inequality each fails with probability less than $1/4$. Fix an outcome satisfying (V1)–(V3).

Step j , for $j = 0, 1, \dots, k-1$. Let E_j be the set of edges of $E[P]$ not used so far. We maintain that $E_j \subseteq E[P_j]$, which holds for $j = 0$. Let F_j be the set of edges of E_j that meet Λ_j .

- (1) For every $w \in \Lambda_j$ use (V2) to *reserve* four edges of O^{res} from w to distinct outlets in $V_{j,0}$, two for each *phase* $c \in \{1, 2\}$. Remove the reserved edges from F_j if they lie in it.
- (2) Give every remaining edge of F_j a phase c such that $V_{j,c}$ contains neither of its ends. This is possible: one end lies in Λ_j , which is disjoint from U_{j+1} , and the other end has only one label.
- (3) For each phase c , decompose the edges of phase c by Lemma 2.1 into at most $|P_j|$ paths, with every vertex an end of at most two of them. At every end $w \in \Lambda_j$ of such a path, append a distinct phase- c reserved edge of w . The result is a trail whose ends lie in

U_{j+1} , all of whose edges meet Λ_j , and which avoids $V_{j,c}$. The path has no repeated vertex, so a vertex can occur twice on the trail only if it is a new end vertex. If some vertex y occurs twice, the part of the trail between its first two occurrences is a closed trail with no other repetition, that is, a cycle; we cut it out. It has length at least three, because the graph is simple and the appended reserved edges are not edges of the path (they were removed from F_j in (1)). Each cut removes an occurrence of a new end vertex, so after at most two cuts we are left with at most two cycles and at most one path P' that joins the two ends of the trail, if these differ. The path P' has length at least two, because each of its edges meets Λ_j and its ends do not.

- (4) For each phase c , join the two ends of every residual path P' by a path through $V_{j,c}$ in $O_{j,c}$, edge-disjointly, using (V1). A vertex of U_{j+1} is an end of at most 2 phase- c paths from the decomposition, and it is the far end of at most one phase- c reserved edge of each w that has it as an outlet. So it lies in at most $b + 2$ pairs. Each connector meets its residual path only at the ends, since its interior lies in $V_{j,c}$, which the residual path avoids. As the residual path has length at least two, the two together form a cycle even if the connector is a single edge.
- (5) Pay for the unused reserved edges.

The edges of E_j that do not meet Λ_j lie in $E[P_{j+1}]$. So after we also remove the connector edges that lie in $E[P]$, the invariant $E_{j+1} \subseteq E[P_{j+1}]$ holds.

No edge is used twice. Step j uses three kinds of edges: edges of $F_j \subseteq E_j$, which are unused by definition; reserved edges wy with $w \in \Lambda_j$ and $y \in U_{j+1}$; and connector edges of colours $(j, 1), (j, 2)$. Every connector edge has both ends in U_{j+1} , because its interior lies in $V_{j,c} \subseteq U_{j+1}$ and so do the ends of residual paths. An earlier step $i < j$ uses only edges that meet Λ_i (its edges of F_i and its reserved edges) and connector edges of colours $(i, \cdot)$. Since Λ_i is disjoint from $U_{i+1} \supseteq U_j$, none of these is a reserved or connector edge of step j . Within step j , the connectors of one phase are edge-disjoint, the two phases use different colours, and connector edges do not meet Λ_j , so they are not in F_j .

The finish and the count. After step $k - 1$ the unused edges of $E[P]$ lie in $E[P_k]$. By Lemma 2.2 and (V3) they cost at most $|P_k|(\log |P_k| + 1) \leq |P_k|(L + 1) \leq 49|P|$ objects. Step j produces at most three cycles per path and pays for at most $4|\Lambda_j|$ reserved edges. So all steps together cost at most $6 \sum_j |P_j| + 4|P| \leq 52|P|$ objects. The total is at most $101|P|$, and O' consists of all reserved edges and of the connector edges that were used. $\square$

5.2 Closing cherries

The second lemma chains cherries into cycles. In the application the ports are the ports of one round, the classes are given by their anchors, and the graphs $R_{Y,j}$ are the closing reservoirs of the anchors. Informally, it says: if every edge to be decomposed has a port end, if edges at a port leave its class, and if each class has K edge-disjoint reservoirs linking through K disjoint slots, then all edges except at most one per pair (non-port, class) can be covered by about $\kappa_Y/2 + 2\Delta$ cycles per class plus $|E'|/(2K)$ cycles in total. Here κ_Y measures how many ports of class Y can share a neighbour.

Lemma 5.2 (Cherries). *Let E' be a set of edges and let Π be a set of vertices, called ports, partitioned into classes Π_Y indexed by a family $\mathcal{Y}$ of vertex sets with $\Pi_Y \subseteq Y$. Let $\Delta, K \geq 1$ be integers. Suppose that:*

- (C1) *every edge of E' has an end in Π , and an edge of E' at a port of Π_Y has its other end outside Y ;*

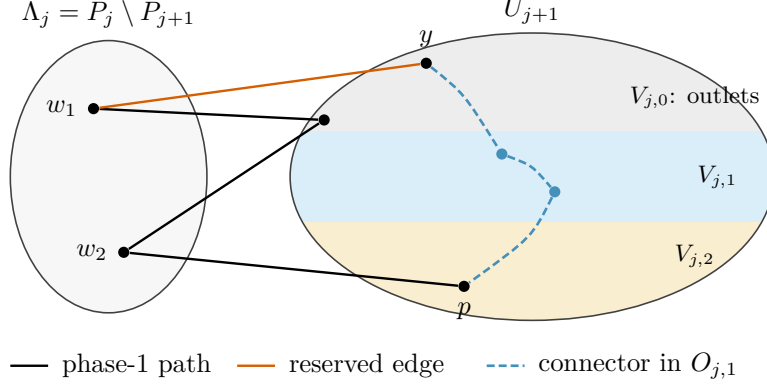


Figure 5.1: One step of the clearing lemma. A phase-1 path of remaining edges at Λ_j avoids $V_{j,1}$. Its end $w_1 \in \Lambda_j$ is extended by a reserved edge to an outlet $y \in V_{j,0}$. Its other end p already lies in P_{j+1} . The two ends, both in U_{j+1} , are joined by a connector through $V_{j,1}$, which closes a cycle. Edges from P_{j+1} to later levels are handled in later steps.

- (C2) every port is incident with at most Δ edges of E' ;
- (C3) for each $Y \in \mathcal{Y}$ there are pairwise disjoint sets $T_{Y,1}, \dots, T_{Y,K} \subseteq Y \setminus \Pi_Y$, the slots, and graphs $R_{Y,1}, \dots, R_{Y,K}$ with $V(R_{Y,j}) \supseteq Y$, the reservoirs, such that $R_{Y,j}$ links through $T_{Y,j}$ with multiplicity Δ ;
- (C4) the edge sets of all graphs $R_{Y,j}$ are pairwise disjoint and disjoint from E' .

For $Y \in \mathcal{Y}$ let $\kappa_Y = \max_x |\{u \in \Pi_Y : xu \in E'\}|$. Then there are $J \subseteq E'$ and a set R' of edges of the graphs $R_{Y,j}$ such that $(E' \setminus J) \cup R'$ decomposes into at most

$$\sum_{Y: \Pi_Y \neq \emptyset} \left(\frac{\kappa_Y}{2} + 2\Delta \right) + \frac{|E'|}{2K}$$

cycles, and for every $x \notin \Pi$ and every Y , at most one edge of J joins x to Π_Y .

Proof. Assign every edge of E' to one of its ends in Π ; edges at non-ports are assigned to their port end. For an edge assigned to $u \in \Pi_Y$, call its other end, which lies outside Y , its *centre*, and call Y its class. For each class Y and centre x , pair up the edges of class Y with centre x into *cherries* uxv with $u, v \in \Pi_Y$. If their number is odd, put one of them into J . Every edge at a non-port x has centre x , so J has the stated property.

Fix a class Y with N_Y cherries. Two cherries *conflict* if they share a vertex. A cherry uxv conflicts with at most $\Delta - 1$ others at each of u and v . Ends lie in Y and centres do not, so a cherry can share u or v only as an end. At the centre, a cherry conflicts with at most $\kappa_Y/2 - 1$ others. So the conflict degree is at most $\kappa_Y/2 + 2\Delta - 3$. Put the cherries greedily into groups of at most K pairwise vertex-disjoint cherries, opening a new group only if every existing group is full or contains a conflicting cherry. At most $\kappa_Y/2 + 2\Delta - 3$ groups contain a conflicting cherry, and fewer than N_Y/K groups are full. So there are at most $\kappa_Y/2 + 2\Delta - 3 + \lceil N_Y/K \rceil$ groups.

Order each group as $p_1x_1q_1, \dots, p_kx_kq_k$, with $k \leq K$, and join q_i to p_{i+1} (indices modulo k) by a path through $T_{Y,i}$ in $R_{Y,i}$ (Figure 5.2). For each slot i , we request these connections for all groups of class Y at once. Each port is an end of at most Δ cherries, so it lies in at most Δ requests for slot i , and (C3) provides edge-disjoint connectors. In a group, the connectors have their interiors in the pairwise disjoint sets $T_{Y,i} \subseteq Y \setminus \Pi_Y$. These sets avoid the ends of the cherries, which are ports of Π_Y , and the centres, which lie outside Y . So each group becomes a cycle; when $k = 1$ the cycle is $p_1x_1q_1$ followed by a connector from q_1 back to

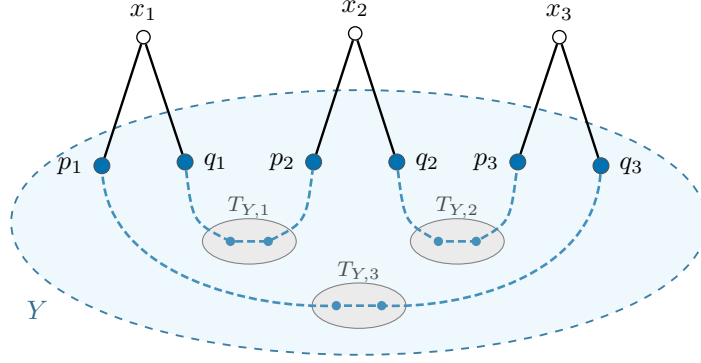


Figure 5.2: Closing a group of three vertex-disjoint cherries $p_i x_i q_i$ whose ends lie in the anchor Y and whose centres lie outside it. The connector from q_i to p_{i+1} runs through its own slot $T_{Y,i}$, so the connectors are internally disjoint from each other and from the cherries, and the result is one cycle.

p_1 . Different groups and different classes use disjoint edges by (C4). Since $\sum_Y N_Y \leq |E'|/2$, summing the number of groups gives the bound. $\square$

5.3 The quotient

The third lemma turns the leftover edges into edges of an auxiliary graph whose cycles lift to cycles of the host graph. Leftovers are of two kinds: *hub edges* au from a hub a to a port u , and *links* uv or uxv between two ports, possibly through a middle vertex x . Every end at a port is extended by a private *donor* edge to a vertex of a pool. A hub edge au with donor w becomes the edge aw of the auxiliary graph, and a link with donors w^- and w^+ at its two ends becomes the edge w^-w^+ (Figure 5.3).

For a cycle of the auxiliary graph to lift to a cycle, the paths represented by its edges must be edge-disjoint and must not pass through each other's interior vertices. The danger is at ports: a port u may lie on up to Δ hub edges and links, and two of the represented paths through u must never lie on a common cycle. So we split the auxiliary graph into vertex-disjoint parts in which every port and every middle lies on at most one represented path. Hub edges are coloured so that hub edges at a common port get distinct colours, and the few hub edges for which this fails are paid for. Links are coloured so that links of one colour share no port or middle. Within one colour, parallel edges are separated into *layers*, so that each layer is simple. The quotient Q is the vertex-disjoint union of all layers. A vertex of the host graph that occurs in several layers thus occurs in Q several times, once per layer; we call these occurrences its *copies*. The size of Q is the total number of copies, and the work is to keep it small. Hubs have few copies because their hub edges are processed in large groups. Pool vertices have few copies because the donors of links are chosen at random, which spreads the links evenly over the pools.

There are two pools, W^- and W^+ , because a link chooses the donor at its tail from W^- and the donor at its head from W^+ . This prevents loops, and it makes the two ends of a link independent, which the occupancy estimate in the proof needs.

Lemma 5.3 (Quotient). *Let $\Delta \geq 2$ and $g \geq 4\Delta + 4$ be integers. Let A, Π, I, W^- and W^+ be pairwise disjoint sets of vertices of a graph Γ (hubs, ports, middles and two pools), and let $W = W^- \cup W^+$. Suppose we are given:*

- *a set $\mathcal{H}$ of hub edges au with $a \in A$ and $u \in \Pi$, and a set $\mathcal{L}$ of links, each an edge uv or a path uxv of Γ with $u \neq v \in \Pi$ and $x \in I$, all edges involved being distinct;*

- every port lies on at most Δ hub edges and links, every middle lies on at most Δ links;
- for every port u , donor sets $D^-(u) \subseteq N_\Gamma(u) \cap W^-$ and $D^+(u) \subseteq N_\Gamma(u) \cap W^+$, each of size at least g ;
- a function $\beta : W \rightarrow \mathbb{Z}_{\geq 0}$ such that for all $a \in A$ and $w \in W$,

$$|\{u : au \in \mathcal{H}, w \in D^-(u) \cup D^+(u)\}| \leq \beta(w). \quad (5.1)$$

Then there are a set $\mathcal{H}_{\text{paid}} \subseteq \mathcal{H}$ and a simple graph Q with

$$|\mathcal{H}_{\text{paid}}| + |V(Q)| \leq |A| + \Delta^3 \sum_{w \in W} \beta(w) + 3\Delta[2 \log \Delta]|W| + 30(|\mathcal{H}| + |\mathcal{L}|) \left(\frac{1}{g} + \frac{1}{\Delta^2} \right),$$

such that whenever Q has a decomposition into k objects, the set $(\mathcal{H} \setminus \mathcal{H}_{\text{paid}}) \cup E(\mathcal{L}) \cup D'$ has a decomposition into at most $2k$ objects for some set D' of donor edges uw with $w \in D^\pm(u)$.

Donor edges are never hub edges or edges of links, because W is disjoint from $A \cup \Pi \cup I$.

Proof. Layers. Let $\mathcal{M}$ be a loopless multigraph, and let $m(v, w)$ be the number of edges between v and w . Its *layers* are the simple graphs $\mathcal{M}_1, \mathcal{M}_2, \dots$, where $\mathcal{M}_i$ contains the i -th edge of each parallel class, and isolated vertices are removed. Their total number of vertices is $\sum_v \max_w m(v, w)$.

The random choices. The proof makes two kinds of random choices. First, every group of hub edges (defined below) gets a colour, independently and uniformly. Then, given the colours, every port u gets a pair of injections ϕ_u^-, ϕ_u^+ , chosen independently and uniformly, whose ranges depend on the colours. All other choices are deterministic functions of these.

Hub edges. For every hub a , split the hub edges at a into groups of size $\lfloor g/2 \rfloor$, except for at most one smaller group. There are at most $|A| + 4|\mathcal{H}|/g$ groups. Colour the groups independently and uniformly with Δ^3 colours, and give each hub edge the colour of its group. Let $\mathcal{H}_{\text{paid}}$ be the set of hub edges au for which another hub edge at u has the same colour. Two hub edges at u come from different hubs, hence from different groups, so $\mathbb{P}(au \in \mathcal{H}_{\text{paid}}) \leq (\Delta - 1)/\Delta^3$ and $\mathbb{E}|\mathcal{H}_{\text{paid}}| \leq |\mathcal{H}|/\Delta^2$. For each hub edge $au \notin \mathcal{H}_{\text{paid}}$ choose a donor $w(au) \in D^-(u) \cup D^+(u)$, by a fixed greedy rule. The donors must be distinct for distinct hub edges at the same port and for hub edges in the same group. This is possible, since fewer than $\Delta + g/2 \leq g$ choices are excluded. For each colour c , let $\mathcal{M}_c$ be the multigraph with an edge $aw(au)$ for every hub edge $au \notin \mathcal{H}_{\text{paid}}$ of colour c . A group contributes at most one edge to any pair (a, w) , so the copies of a in the layers of all $\mathcal{M}_c$ number at most the number of groups at a . Distinct hub edges at a with the same donor w have distinct ports, so by (5.1) every pool vertex w has at most $\beta(w)$ copies per colour. These layers therefore have at most $|A| + 4|\mathcal{H}|/g + \Delta^3 \sum_w \beta(w)$ vertices.

Links. Colour the links greedily with at most 3Δ colours so that links of the same colour share no port and no middle; a link meets at most $3(\Delta - 1)$ others. Orient every link from one end port to the other. For every port u , let ϕ_u^- be a uniformly random injection from the links leaving u into $D^-(u)$ minus the donors of hub edges at u , and let ϕ_u^+ be a uniformly random injection from the links entering u into $D^+(u)$ minus those donors. The domains have size at most Δ , and the ranges have size at least $g - \Delta \geq g/2$. A link from u to v thus gets a *tail junction* $w^- = \phi_u^-(\text{link}) \in W^-$ and a *head junction* $w^+ = \phi_v^+(\text{link}) \in W^+$. For each colour c , let $\mathcal{N}_c$ be the multigraph with an edge w^-w^+ for every link of colour c , and split it into layers.

To bound the number of vertices we use an occupancy estimate. Let $\xi_1, \dots, \xi_q$ be independent random variables, each taking every value with probability at most π , and let $X_y = |\{i : \xi_i = y\}|$. Then for every integer $t \geq 1$

$$\mathbb{E} \max_y X_y \leq t + 2e\pi q + 4e2^{-t}q. \quad (5.2)$$

Indeed, $\mathbb{P}(X_y \geq j) \leq \mathbb{E}(\binom{X_y}{j}) \leq (\mathbb{E}X_y)^j/j! \leq (e\mathbb{E}X_y/j)^j$. For $j \geq T = \max(t, \lceil 2e\pi q \rceil)$ we have $e\mathbb{E}X_y/j \leq 1/2$, so this is at most $(e\mathbb{E}X_y/j)2^{1-j}$, and summing gives $\mathbb{E}[X_y \mathbf{1}_{X_y \geq T}] \leq 4e2^{-T}\mathbb{E}X_y$. Now use $\max_y X_y \leq T - 1 + \sum_y X_y \mathbf{1}_{X_y \geq T}$ and $\sum_y \mathbb{E}X_y = q$.

Condition on the colours of the groups, which determine the hub donors and hence the ranges of all injections, and on all the injections ϕ_u^- , which determine the tail junctions. The injections ϕ_v^+ are still independent and uniform. Fix a colour c and a pool vertex $w \in W^-$. The links of colour c with tail junction w have distinct head ports, because links of one colour share no port. So their head junctions are determined by different injections ϕ_v^+ . They are therefore independent, and each is uniform on a set of size at least $g/2$. The number of copies of w in the layers of $\mathcal{N}_c$ is the maximum multiplicity of an edge at w . By (5.2) with $t = \lceil 2 \log \Delta \rceil$ and $\pi = 2/g$, its conditional expectation is at most $t + 4eq_w/g + 4eq_w/\Delta^2$, where q_w is the number of these links. The same holds for $w \in W^+$, conditioning on the head junctions instead. Summing over w and over the 3Δ colours, and using that every link is counted once at its tail junction and once at its head junction, the expected number of vertices in the layers of all $\mathcal{N}_c$ is at most $3\Delta t|W| + 8e|\mathcal{L}|/g + 8e|\mathcal{L}|/\Delta^2$.

The graph Q . Let Q be the vertex-disjoint union of all layers of all $\mathcal{M}_c$ and $\mathcal{N}_c$. Adding up, $\mathbb{E}(|\mathcal{H}_{\text{paid}}| + |V(Q)|)$ is at most the bound in the lemma, since $8e < 30$. Fix an outcome of the random choices that achieves this bound.

Lifting. An edge aw of Q that comes from the hub edge au represents the path auw , and an edge w^-w^+ that comes from a link represents the path w^-uvw^+ or w^-uxvw^+ . These paths have three properties.

- (L1) Distinct edges of Q represent edge-disjoint paths. The hub edges and links are distinct, and the donor edges at a port are distinct by construction: hub donors are distinct, and the injections avoid them and map into the disjoint sets W^- and W^+ .
- (L2) In each layer, distinct vertices are distinct vertices of Γ . These are hubs and pool vertices.
- (L3) In each layer, the interior vertices of the represented paths are pairwise distinct and are not vertices of the layer. In one colour a port lies on at most one surviving hub edge, links of one colour share no port or middle, and ports and middles are neither hubs nor pool vertices.

Take a decomposition of Q into k objects. A cycle of Q lies in one layer, and by (L2) and (L3) replacing each of its edges by the represented path gives a cycle of Γ . For a single edge of Q we output only its *core*: the hub edge, or the one or two edges of the link. Its donor edges are left unused. By (L1) this gives at most $2k$ edge-disjoint objects covering exactly $(\mathcal{H} \setminus \mathcal{H}_{\text{paid}}) \cup E(\mathcal{L}) \cup D'$, where D' is the set of donor edges on lifted cycles. $\square$

6 Proof of the reduction theorem

Let G be a graph on n vertices. If $2e(G)/n < D$ we pay for all edges, and $f(G) < Dn/2$. Otherwise we build the hierarchy of Section 3.3, with rounds $1, \dots, R$, and use the notation of Section 3.

6.1 Plan

The proof has three parts.

Random preparation (Section 6.2). One random experiment, carried out once for all pieces, does three things. It colours the edges of every piece Y , so that Y keeps half of its edges

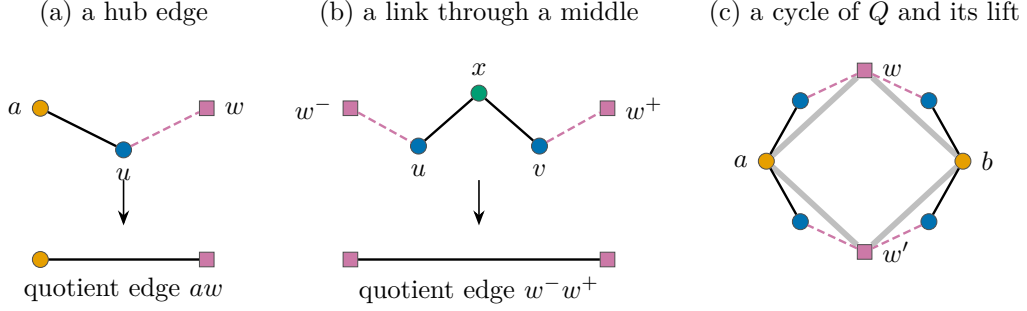


Figure 5.3: The quotient. (a) A hub edge au , extended by a donor edge uw (dashed) to a pool vertex, becomes the edge aw of Q . (b) A link uxv between two ports becomes an edge between two pool vertices. (c) The cycle $abw'w$ of Q (grey) lifts to an 8-cycle of G . The branch vertices are hubs and pool vertices, and the private ports in between keep the lift simple.

for itself and divides the rest into small random reservoirs for the later rounds. It chooses random *slots* inside Y , through which the connectors that close cherries will run. And it chooses sparse random *pools* of vertices for the quotients. For a few ports these choices fail: their anchor is *bad*, or they fall into a slot of their anchor (these ports are *lost*), or they have too few donor edges to the pools (these ports are *deficient*).

Assembly (Section 6.3). For every outcome of the random preparation, we decompose the rounds in the order $R, R-1, \dots, 1$. Round ℓ starts from the edges assigned to its pieces, minus the reservoir edges that later rounds have already used, and runs the four steps of Section 1.1; Figure 6.1 in Section 6.3 gives an overview. Lost ports are cleared together with hubs and fresh vertices, and the edges at deficient ports are paid for. Lemma 6.5 checks that every edge of G is used exactly once.

Counting (Section 6.4). The cost of the assembly depends on the random outcome only through the lost and deficient ports and the pools. We collect this dependence in one random variable Ω , show that $\mathbb{E}\Omega = o_D(n)$, and fix an outcome with $\Omega \leq \mathbb{E}\Omega$. Adding up all costs then proves Theorem 1.2.

Where the costs go. As in the budget of Section 1.1, only three things cost $\Theta(n)$: the edges of the final sparse graph G_{R+1} ; the fresh vertices, which are cleared and paired at most twice each; and the lifted decompositions of the quotients, which have $o_D(n)$ vertices in total and are handled by induction. Everything else (long cycles, hubs, cherries, unlucky ports and pools) is summable over the rounds by Lemma 3.5(c), so it costs $o_D(n)$.

6.2 Random preparation

Reservoirs and slots. Let Y be a piece of round r . It lends to the rounds $\ell \in \{r+2, \dots, R\}$. For each such round it provides $K = M_\ell^2$ *closing reservoirs*, one for each position of a cherry in a chain of Lemma 5.2, and one *donor reservoir* for the quotient. Each closing reservoir comes with its own random set of *slot* vertices through which its connectors run. The numbers are chosen as follows. With $K = M_\ell^2$ closing reservoirs, Lemma 5.2 costs only $|E'|/(2K) \leq n/(2M_\ell)$ beyond the crowding terms. Each slot set is a random set of density M_ℓ^{-4} . This is sparse enough that a given port lies in some slot with probability only $M_\ell^2 \cdot M_\ell^{-4} = M_\ell^{-2}$, and dense enough for the linking lemma, because the reservoirs have

degrees about λ_r^{46} , far larger than any fixed power of M_ℓ . Put

$$\chi_r = \sum_{\ell=r+2}^R (M_\ell^2 + 1),$$

the number of reservoirs of Y . By Lemma 3.5 we have $M_\ell \leq M_{r+2}2^{-(\ell-r-2)}$ and $M_{r+2} \leq (55\mu_r)^{111}$, so $\chi_r \leq 3(55\mu_r)^{222} \leq \lambda_r^4/2$.

Definition 6.1 (Reservoirs). Colour every edge of H_Y independently at random. It gets the *own* colour with probability $1/2$. Each of the following colours has probability $1/(2\chi_r)$: for every $\ell \in \{r+2, \dots, R\}$, the *closing* colours (ℓ, j) with $j \in [M_\ell^2]$, and the *donor* colour ℓ . If $r+2 > R$, all edges get the own colour. The *own graph* O_Y , the *closing reservoirs* $R_{Y,\ell,j}$ and the *donor reservoirs* $R_{Y,\ell}^{\text{don}}$ are the spanning subgraphs of H_Y formed by these colour classes; the last two kinds are the *reservoirs* of Y . Independently, for every ℓ , give every vertex of Y a *slot* for round ℓ . It is slot j with probability M_ℓ^{-4} for each $j \in [M_\ell^2]$, and no slot otherwise. Let $T_{Y,\ell,j}$ be the set of vertices with slot j for round ℓ . The piece Y is *good* if

(G1) O_Y is an $(s_r/4)$ -expander, and

(G2) every $R_{Y,\ell,j}$ links through $T_{Y,\ell,j}$ with multiplicity M_ℓ .

Otherwise it is *bad*.

Lemma 6.2. *Every piece Y is bad with probability at most $|Y|^{-2}$.*

Proof. Let $N = |Y|$, so $\lambda_r^{54} \leq N \leq M_r$ and $\log N \leq 3\lambda_r$. By Lemma 4.2 with $p = 1/2$, (G1) fails with probability at most N^{-10} . Each $R_{Y,\ell,j}$ is a random subgraph with $p = 1/(2\chi_r) \geq \lambda_r^{-4}$, so by Lemma 4.2 it is a $(\lambda_r^{46}/2)$ -expander with probability at least $1 - N^{-10}$. The set $T_{Y,\ell,j}$ is M_ℓ^{-4} -random and independent of the colouring. Lemma 4.3 with $t = M_\ell$ and $\rho = M_\ell^{-4}$ needs $\lambda_r^{46}/2 \geq C_* M_\ell^{13} (3\lambda_r)^{18}$, which holds since $M_\ell \leq (55\mu_r)^{111}$. So (G2) fails for a given (ℓ, j) with probability at most $N^{-10} + N^{-3}$. There are at most $\chi_r \leq \lambda_r^4 \leq N^{1/10}$ pairs (ℓ, j) , so the total failure probability is at most N^{-2} . $\square$

Definition 6.3 (Lost and live ports). A port u of round ℓ with anchor $Y = \text{anc}_\ell(u)$ is *lost* if Y is bad or u lies in some slot set $T_{Y,\ell,j}$. Otherwise it is *live*. Let Lost_ℓ be the set of lost ports of round ℓ .

So a live port avoids all slot sets of its anchor for its round, as Lemma 5.2 requires, and its anchor is good. If a piece is bad, every port anchored at it in every round is lost, so a bad piece never lends anything.

Pools. Independently of all previous choices, give every vertex of G at most one *pool label* (ℓ, r, σ) , with $3 \leq \ell \leq R$, $1 \leq r \leq \ell - 2$ and $\sigma \in \{-, +\}$. Each label has probability $q_\ell 2^{-(\ell-r)}$, where $q_\ell = M_\ell^{-5}$. The total probability is at most $\sum_\ell q_\ell < 1$. Let $W_{\ell,r,\sigma}$ be the set of vertices with label (ℓ, r, σ) , let $W_\ell^\sigma = \bigcup_r W_{\ell,r,\sigma}$ and $W_\ell = W_\ell^- \cup W_\ell^+$, and for $w \in W_\ell$ let $r(w)$ be the second coordinate of its label. Every vertex lies in W_ℓ with probability less than q_ℓ . So the pools are very sparse, which keeps the quotient small. Yet every port still has many donor candidates, because its anchor gives it about λ_r^{46} edges of the donor reservoir (Lemma 6.4).

The pool of round ℓ is split by the round r of the anchors it serves. The reason is condition (5.1) of Lemma 5.3: a pool vertex w may serve as a donor for ports with many different anchors, but only anchors of one round $r(w)$, and these anchors all contain w . The weights $2^{-(\ell-r)}$ make the resulting cost summable over r (Lemma 6.6).

Donors. Let u be a port of round ℓ with anchor Y of round r . Its donor sets are

$$D^\sigma(u) = N_{R_{Y,\ell}^{\text{don}}}(u) \cap W_{\ell,r,\sigma} \quad (\sigma \in \{-, +\}).$$

Let $g_\ell = M_\ell^{10}$. The port u is *deficient* if $|D^-(u)| < g_\ell$ or $|D^+(u)| < g_\ell$. Let Def_ℓ be the set of deficient ports of round ℓ .

Lemma 6.4. *Every port of round ℓ is deficient with probability at most $2e^{-g_\ell}$.*

Proof. For a fixed σ , $|D^\sigma(u)|$ is a sum of independent indicators, one for each of the more than s_r neighbours w of u in H_Y : the edge uw has the donor colour ℓ , and $w \in W_{\ell,r,\sigma}$. By Lemma 3.5(b), $2^{\ell-r} \leq \lambda_r$ and $M_\ell \leq (55\mu_r)^{111}$, so the mean is at least

$$\lambda_r^{50} \cdot \lambda_r^{-4} \cdot M_\ell^{-5} 2^{-(\ell-r)} \geq \lambda_r^{45} M_\ell^{-5} \geq 4M_\ell^{10} = 4g_\ell.$$

By Lemma 2.3 it is less than g_ℓ with probability at most e^{-g_ℓ} . $\square$

Table 6.1 summarises the random choices. They are all independent, and they are made after the hierarchy and the anchors have been fixed. Until Section 6.4 the outcome is arbitrary: the assembly works for every outcome, and only its cost depends on it.

object	distribution	purpose
own graph $O_Y \subseteq H_Y$	each edge with probability $\frac{1}{2}$	clearing inside Y (Lemma 5.1)
closing reservoirs $R_{Y,\ell,j}$, $j \in [M_\ell^2]$	each edge with probability $\frac{1}{2\chi_r}$	closing cherries in round ℓ (Lemma 5.2)
donor reservoir $R_{Y,\ell}^{\text{don}}$	each edge with probability $\frac{1}{2\chi_r}$	donor edges in round ℓ (Lemma 5.3)
slot sets $T_{Y,\ell,j} \subseteq Y$	each vertex with probability M_ℓ^{-4}	interiors of cherry connectors
pools $W_{\ell,r,\pm} \subseteq V(G)$	each vertex with probability $q_\ell 2^{-(\ell-r)}$	vertices of the quotient Q_ℓ

Table 6.1: The random choices for a piece Y of round r and a later round $\ell \geq r + 2$, and for the pools. Lost and deficient ports are the ports for which these choices fail.

6.3 Decomposing the rounds in reverse order

Fix any outcome of the random preparation. We process the rounds $\ell = R, R - 1, \dots, 1$ in this order, and keep a set Used of reservoir edges that the rounds processed so far have used. It is initially empty. Let $\mathbf{E}_\ell = \bigcup_{X \in \mathcal{Y}_\ell} E_X$ be the set of edges assigned in round ℓ . A vertex that is not a hub of round ℓ has fewer than M_ℓ edges in $\mathbf{E}_\ell$, because all of them lie in its only piece, which has at most M_ℓ vertices. Round ℓ consists of the following four steps.

Step 1: clear. For every piece X of round ℓ , let $F_X = E_X \setminus \text{Used}$, and let

$$P_X = X \cap (\text{Hub}_\ell \cup \text{Fresh}_\ell \cup \text{Lost}_\ell).$$

Let O_X be the own graph of X if X is good, and $O_X = H_X$ if X is bad. Apply Lemma 5.1 to $O = O_X$, $E = F_X$ and $P = P_X$, and let $E'_X = F_X \setminus (F_X[P_X] \cup O')$ be the set of edges of F_X that remain.

Legitimacy. We need $O_X \subseteq F_X$, that is, no edge of O_X has been used by a later round, so that the objects produced use only edges of F_X . Later rounds use only reservoir edges.

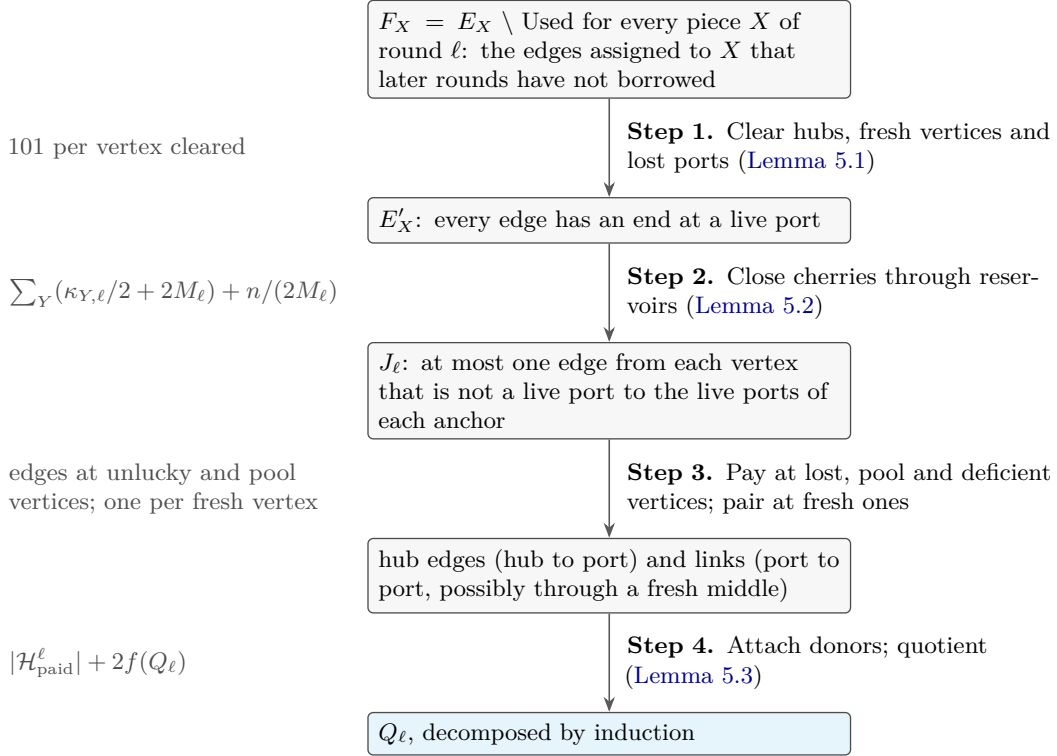


Figure 6.1: One round of the assembly. Boxes show the edges that remain after each step. The text to the right of each arrow says what the step does, and the text to the left says what it costs.

If X is good, O_X consists of own-coloured edges. If X is bad, every port anchored at X is lost, so no cherry of class X is ever closed and no donor set of X is ever used; hence no edge of X is in Used . Moreover O_X is an $(s_\ell/4)$ -expander with $s_\ell/4 \geq (3\lambda_\ell)^{40} \geq \log^{40} |X|$, and $|X| \geq m_\ell \geq N_0$.

Outcome. Every vertex of X is a hub, a fresh vertex, a lost port or a live port, so every edge of E'_X has an end at a live port. For $\ell \leq 2$ there are no ports, so $P_X = X$, $E'_X = \emptyset$, and the round ends here.

Cost. At most $101|P_X|$ objects.

Step 2: close cherries ($\ell \geq 3$). Let $E' = \bigcup_X E'_X$. Apply Lemma 5.2 to E' , with the live ports of round ℓ as ports, classified by their anchors, and with $\Delta = M_\ell$, $K = M_\ell^2$, $T_{Y,j} = T_{Y,\ell,j}$ and $R_{Y,j} = R_{Y,\ell,j}$. Add the reservoir edges used to Used , and let J_ℓ be the leftover set.

Legitimacy. (C1): Let u be a live port with anchor Y of round r , and let $uy \in E' \subseteq E(G_\ell)$. By the separation property (H6), $y \notin Y$. (C2): u is not a hub, so it has fewer than M_ℓ edges in E_ℓ . (C3): The anchor Y of a live port is good, so the closing reservoirs $R_{Y,\ell,j}$, which span Y , link through $T_{Y,\ell,j}$ by (G2). The sets $T_{Y,\ell,j}$ are pairwise disjoint and avoid the live ports anchored at Y . The colours (ℓ, j) are used only in round ℓ , so none of these edges is in Used . (C4): The graphs H_Y are pairwise edge-disjoint, and they are disjoint from $E' \subseteq E_\ell$.

Outcome. J_ℓ has at most one edge between each vertex that is not a live port and the live ports of each anchor.

Cost. Every edge of E' has a live port as an end, so $|E'| \leq nM_\ell$. Since $\kappa_Y \leq \kappa_{Y,\ell}$, the number of cycles is at most

$$\sum_Y \left(\frac{\kappa_{Y,\ell}}{2} + 2M_\ell \right) + \frac{n}{2M_\ell},$$

where Y runs over the ν_ℓ pieces of rounds at most $\ell - 2$.

Step 3: pay and pair. Some leftover edges cannot enter the quotient: edges at lost ports, which have no usable anchor; edges at deficient ports, which have too few donors; and edges at pool vertices, which are reserved as vertices of the quotient and must not also play another role. Pay for every edge of J_ℓ with an end in $\text{Lost}_\ell \cup W_\ell \cup \text{Def}_\ell$. Then, at every fresh vertex $x \notin W_\ell$, pair up the remaining edges of J_ℓ at x into paths uxv , and pay for the leftover edge if their number is odd. Since the two edges at x are distinct edges of a simple graph, $u \neq v$.

Cost. For a hub h of round ℓ , let $c_\ell(h)$ be the number of anchors Y such that some edge of $\mathbf{E}_\ell$ joins h to a port with anchor Y , and put

$$\omega_\ell(v) = \begin{cases} c_\ell(v) & \text{if } v \in \text{Hub}_\ell, \\ M_\ell & \text{otherwise.} \end{cases}$$

These weights depend only on the hierarchy, not on the random choices. A vertex that is not a hub has J_ℓ -degree less than M_ℓ . Every edge of J_ℓ at a hub h ends at a live port, because every edge of E' has a live-port end and hubs are not ports. So by the outcome of Step 2, h has J_ℓ -degree at most $c_\ell(h)$. Hence the number of edges paid for in this step is at most

$$M_\ell(|\text{Lost}_\ell| + |\text{Def}_\ell|) + \sum_{v \in W_\ell} \omega_\ell(v) + |\text{Fresh}_\ell|. \quad (6.1)$$

Step 4: quotient. For a vertex w and a round r , let $\beta_r(w)$ be the number of pieces of round r that contain w . Apply Lemma 5.3 with $\Gamma = G$, $\Delta = M_\ell$ and $g = g_\ell$. The hubs are $A = \text{Hub}_\ell \setminus W_\ell$, the ports Π are the live ports outside $W_\ell \cup \text{Def}_\ell$, the middles are $I = \text{Fresh}_\ell \setminus W_\ell$, the pools are $W^\pm = W_\ell^\pm$, and the donor sets are $D^\pm(u)$. The hub edges are the remaining edges of J_ℓ between A and Π . The links are the remaining edges of J_ℓ inside Π , together with the paths formed in Step 3. Finally, $\beta(w) = \beta_{r(w)}(w)$. Pay for the edges of $\mathcal{H}_{\text{paid}}^\ell$, the set $\mathcal{H}_{\text{paid}}$ given by the lemma. Take a decomposition of Q_ℓ into $f(Q_\ell)$ objects, lift it, and add the donor edges used to Used .

Legitimacy. The sets A , Π , I , W^- and W^+ are pairwise disjoint, since hubs, ports and fresh vertices are different roles and the pools were removed from the first three. After Step 3, every edge of J_ℓ has a port in Π as one end. Its other end is a hub in A , a port in Π , or a fresh vertex in I , because all other cases were paid for in Step 3. So the remaining edges are exactly the hub edges and the edges of links. Ports and middles are not hubs, so ports lie on fewer than M_ℓ hub edges and links, and middles on fewer than $M_\ell/2$ links. Each hub edge contains one edge of $\mathbf{E}_\ell$ at a port and each link contains two, so $|\mathcal{H}| + |\mathcal{L}| \leq nM_\ell$. The donor sets have size at least $g_\ell \geq 4M_\ell + 4$, because the ports in Π are not deficient, and the donor colour ℓ is used only in round ℓ . For condition (5.1), let au be a hub edge and $w \in D^\pm(u)$. Then the anchor of u is a piece of round $r(w)$ that contains w . Distinct such ports u have distinct anchors, because J_ℓ has at most one edge from a to the live ports of each anchor. So there are at most $\beta_{r(w)}(w)$ of them.

Cost. At most $|\mathcal{H}_{\text{paid}}^\ell| + 2f(Q_\ell)$ objects. With $t_\ell = \lceil 2 \log M_\ell \rceil$ and $|\mathcal{H}| + |\mathcal{L}| \leq nM_\ell$, the lemma gives

$$|\mathcal{H}_{\text{paid}}^\ell| + |V(Q_\ell)| \leq |\text{Hub}_\ell| + M_\ell^3 \sum_{w \in W_\ell} \beta_{r(w)}(w) + 3M_\ell t_\ell |W_\ell| + 30n \left(\frac{1}{M_\ell^9} + \frac{1}{M_\ell} \right). \quad (6.2)$$

Lemma 6.5 (Every edge is used exactly once). *The long cycles, the single edges of G_{R+1} , and the objects produced in Steps 1–4 of all rounds form a decomposition of $E(G)$.*

Proof. By Lemma 3.4, $E(G)$ is partitioned into the long cycles, the edges of G_{R+1} , and the sets E_X for all pieces X . Fix a piece X of round ℓ . Every edge of E_X in a reservoir colour is used by at most one object, namely in its target round $\ell' \geq \ell + 2$, which is processed before round ℓ ; when this happens it is added to Used. Every other edge of E_X , and every reservoir edge of X that was not used, lies in F_X when round ℓ starts. Within round ℓ , the edges of F_X are used exactly once: Step 1 uses $F_X[P_X] \cup O'$, and passes the rest, E'_X , to Step 2; Step 2 uses $E' \setminus J_\ell$ and passes J_ℓ on; Step 3 pays for some edges of J_ℓ and passes the rest on as hub edges and links; and Step 4 covers these exactly once by Lemma 5.3. The other edges used in round ℓ are reservoir edges of earlier pieces, in colours (ℓ, j) and ℓ ; each of them is used at most once, by Lemmas 5.2 and 5.3, and is then added to Used, so it is not used again when its owner is processed. $\square$

6.4 Counting

The random cost. The costs of Steps 1, 3 and 4 depend on the random outcome through the lost and deficient ports and the pools. We collect these terms in the *random cost*

$$\Omega = \sum_{\ell=3}^R \left[\underbrace{(101 + M_\ell)|\text{Lost}_\ell|}_{\text{lost ports}} + \underbrace{\sum_{v \in W_\ell} \omega_\ell(v)}_{\text{pool vertices}} + \underbrace{M_\ell|\text{Def}_\ell|}_{\text{deficient ports}} + \underbrace{M_\ell^3 \sum_{w \in W_\ell} \beta_{r(w)}(w) + 3M_\ell t_\ell |W_\ell|}_{\text{size of } Q_\ell} \right].$$

The first term pays for clearing lost ports (Step 1) and for their remaining edges (Step 3). The second and third pay for the edges at pool vertices and at deficient ports (Step 3). The last two bound the size of the quotient (Step 4).

The weights ω_ℓ in the second term are what make it easy to estimate. The edges paid for at a pool vertex are edges of J_ℓ , and J_ℓ depends on the random choices, including the pools of later rounds through the set Used. So the degrees of J_ℓ at pool vertices are not independent of the pools. But they are bounded by the weights ω_ℓ , which depend only on the hierarchy, and a sum of fixed weights over a random set is easy to control.

Lemma 6.6. $\mathbb{E}\Omega = o_D(n)$.

Proof. Fix $\ell \geq 3$. A port u with anchor Y of round $r \leq \ell - 2$ has $|Y| \geq m_r \geq M_\ell$, so $\mathbb{P}(u \in \text{Lost}_\ell) \leq |Y|^{-2} + M_\ell^2 \cdot M_\ell^{-4} \leq 2M_\ell^{-2}$ by Lemma 6.2. Hence $\mathbb{E}(101 + M_\ell)|\text{Lost}_\ell| \leq 4n/M_\ell$. Every pair (h, Y) counted by $c_\ell(h)$ is witnessed by a distinct edge of $\mathbf{E}_\ell$ at a port, so $\sum_v \omega_\ell(v) \leq 2nM_\ell$, and therefore $\mathbb{E} \sum_{v \in W_\ell} \omega_\ell(v) \leq 2n/M_\ell^4$. By Lemma 6.4, $\mathbb{E}M_\ell|\text{Def}_\ell| \leq 2nM_\ell e^{-g_\ell} \leq n/M_\ell$. Since $\sum_w \beta_r(w) = \sum_{Y \in \mathcal{Y}_r} |Y| < 4n/3$,

$$\mathbb{E} \sum_{w \in W_\ell} \beta_{r(w)}(w) = \sum_{r \leq \ell-2} 2q_\ell 2^{-(\ell-r)} \sum_w \beta_r(w) \leq \frac{4}{3} n q_\ell,$$

so the fourth term has expectation at most $2n/M_\ell^2$. The last term has expectation at most $3M_\ell t_\ell q_\ell n \leq n/M_\ell$. Altogether $\mathbb{E}\Omega \leq \sum_\ell 10n/M_\ell \leq 20n/D^2$ by Lemma 3.5(a). $\square$

We now fix an outcome of the random preparation with $\Omega \leq \mathbb{E}\Omega$, and carry out the assembly of Section 6.3 for it. Table 6.2 lists all objects produced. Clearing costs at most $101 \sum_\ell \sum_X |P_X|$, and

$$\sum_X |P_X| \leq \sum_X |X \cap \text{Hub}_\ell| + |\text{Fresh}_\ell| + |\text{Lost}_\ell|,$$

since fresh and lost vertices lie in only one piece. The other entries follow from the costs of Steps 2–4, Lemma 3.7, $\Omega = o_D(n)$ and Lemma 3.5(c).

objects	number	reason
long cycles $\mathcal{C}_r$	$o_D(n)$	Lemma 3.7(e)
single edges of G_{R+1}	$< Dn/2$	Lemma 3.4
clearing hubs	$o_D(n)$	Lemma 3.7(b)
clearing fresh vertices	$\leq 202n$	Lemma 3.7(a)
clearing lost ports; paying at lost, pool and deficient vertices	$\leq \Omega = o_D(n)$	Steps 1 and 3, Lemma 6.6
cherry cycles	$o_D(n)$	Lemma 3.7(c),(d) and Lemma 3.5(c)
unpaired edges at fresh vertices	$\leq 2n$	Lemma 3.7(a)
paid hub edges in the quotients	$o_D(n)$	(6.2), see below
lifted decompositions	$\leq 2 \sum_{\ell} f(Q_{\ell})$	Lemma 5.3

Table 6.2: The cost of the decomposition. Only the final sparse graph, the fresh vertices (at most twice each) and the lifted quotients cost $\Theta(n)$, and the quotients have only $o_D(n)$ vertices in total. Everything else is summable over the rounds.

Summing (6.2) over ℓ , the first term is $o_D(n)$ by [Lemma 3.7\(b\)](#), the next two are the last two terms of Ω , and the last is $o_D(n)$ by [Lemma 3.5\(c\)](#). So

$$\sum_{\ell} (|\mathcal{H}_{\text{paid}}^{\ell}| + |V(Q_{\ell})|) = o_D(n). \quad (6.3)$$

Proof of Theorem 1.2. Choose D so large that all requirements in [Table 6.3](#) hold, that $m_R \geq N_0$ for the constant N_0 of [Lemma 5.1](#), and that the left side of (6.3), together with all entries $o_D(n)$ of [Table 6.2](#), is at most $n/4$. These conditions involve only the constant C_* of [Lemma 4.3](#), the constant N_0 , and the absolute constants in [Table 6.3](#), so D is an absolute constant. By [Lemma 6.5](#) the objects produced form a decomposition of $E(G)$, so $\sum_{\ell} |V(Q_{\ell})| \leq n/4$ and

$$f(G) \leq \left(\frac{D}{2} + 202 + 2 + \frac{1}{4} \right) n + 2 \sum_{\ell} f(Q_{\ell}).$$

So [Theorem 1.2](#) holds with $C_0 = D/2 + 205$. □

6.5 The choice of exponents

The exponents in $s = \lceil \lambda^{50} \rceil$, $\tau = \lceil \lambda^{53} \rceil$ and $m = \lceil \lambda^{54} \rceil$ are not optimised. [Table 6.3](#) collects everything the proof needs from them. In short: the pieces must be robust enough to lend edges to later rounds and to clear vertices (s large); the cut threshold must be large enough for the splitting step but negligible compared with the order of the pieces ($s \log^2 M \ll \tau \ll m$); and the next degree, about $s\lambda + m$, must still be polylogarithmic in d .

7 Sources

Every ingredient of the proof is listed here with its source. Numbers of results in [4] refer to its arXiv version.

- *Classical tools.* [Lemma 2.2](#) is the classical $O(n \log n)$ argument. [Lemma 2.1](#) is Lovász's theorem [17] in the form of [4, Corollary 22]. [Theorem A.1](#) is Haxell's theorem [11]. The probabilistic tools are standard [12, 8, 18], and Hall's theorem [9] is used in [Lemma 5.1](#).

requirement	where it is used
$\tau_r \geq 256s_r \log^2 M_r$	splitting steps of round r are legal (Lemma 3.2)
$60\lambda_r^{51} + \frac{4}{3}m_r \leq \lambda_r^{55}$	the next degree is $d_{r+1} \leq \lambda_r^{55}$, see (H5)
$\frac{4}{3}\tau_r/m_r \leq 2/\lambda_r$ and $\log m_r \geq 54\mu_r$	hubs and crowding are summable over rounds (Lemma 3.7)
$m_r \geq 3\lambda_r M_\ell$	anchors are few (Lemma 3.7(d)), and bad anchors are rare compared with slots (Lemma 6.6)
$s_r/4 \geq (3\lambda_r)^{40}$	a piece can clear vertices with its own edges (Lemma 5.1)
$3(55\mu_r)^{222} \leq \lambda_r^4/2$	there are few reservoirs, $\chi_r \leq \lambda_r^4/2$ (Section 6.2)
$s_r/2 \geq 100 \log M_r$, $m_r \geq \lambda_r^{40}$ and $\lambda_r^{46}/2 \geq C_* M_\ell^{13} (3\lambda_r)^{18}$	own graphs are expanders, and reservoirs link through sparse slots (Lemma 6.2)
$\lambda_r^{45} \geq 4M_\ell^{15}$	donor sets are large (Lemma 6.4)

Table 6.3: What the proof needs from the exponents, for all rounds r and $\ell \geq r + 2$. By Lemma 3.5, $M_\ell \leq (55\mu_r)^{111}$, and $\log M_r \leq 2\lambda_r + 1$. All requirements hold if D is large, those involving M_ℓ because $\lambda_r = 2^{\mu_r}$ is exponentially larger than any fixed power of μ_r .

- *Expanders.* Sublinear expanders are due to Komlós and Szemerédi [13, 14], and their robust form to Haslegrave, Kim and Liu [10] and Sudakov and Tomon [20]. Definition 2.4 is [4, Definition 11].
- *Rounds.* The overall scheme of rounds with long-cycle removal, splitting into expanders and polylogarithmic decay of the degree is that of Bucić and Montgomery [4], including the splitting step ([4, Lemma 14]) and the use of long cycles in expanders ([4, Lemma 25]). The depth-first-search proof of Lemma 3.1 is a standard argument (see [15]) and is shorter than the one in [4].
- *Thin cuts.* The rule that moves high-degree vertices into the separator (Lemma 3.2(iii), Lemma 3.3(d)) and its use to bound crowding (Lemma 3.7(c)) are due to Coffey [5].
- *Linking.* Lemma 4.3 extends [4, Theorem 16] to sparse random sets and large multiplicities. Its proof in Appendix A follows [4]: Claims A.3 and A.8 are [4, Propositions 8 and 18], Claim A.7 follows [4, Lemma 17], and the amplification by colouring follows [4, Theorem 16]. The certificate of Claim A.5 simplifies [4, Proposition 13], and in Claim A.4, which follows [4, Lemma 9], Haxell’s theorem replaces the theorem of Aharoni and Haxell [1] used there. Lemma 4.2 replaces [4, Lemma 15] and has a standard proof.
- *Anchors, reservoirs, cherries and the quotient.* The idea of lending random colour classes of pieces at least two rounds back, of closing cherries through them, of compressing the leftovers into a quotient graph through private edges to random pools, and of finishing by induction on the number of vertices, is due to Coffey [5]. So is the use of a vortex, in the sense of iterative absorption [2], to clear vertices locally (Lemma 5.1).
- *Simplifications in this exposition.* Compared with [5], the present account uses one kind of piece, kept whole, instead of several kinds of parts; a single clearing lemma; the greedy grouping of Lemma 5.2 instead of a routing scheme for cherry systems; a simplified quotient with a single averaging argument (Lemma 5.3); the growth certificate in the proof of Lemma 4.3; and smaller, explicitly motivated parameters (Table 6.3).

A Proof of the linking lemma

This appendix proves [Lemma 4.3](#), following the outline given after its statement in [Section 4](#).

Two tools

We use two further tools. The first is Haxell's condition for disjoint representatives, which turns one-path-at-a-time statements into many-paths-at-once statements.

Theorem A.1 (Haxell [11]). *Let $h \geq 1$ and let $\mathcal{A}_1, \dots, \mathcal{A}_q$ be families of sets of size at most h . Suppose that for every nonempty $I \subseteq [q]$ and every set S with $|S| \leq (2h - 1)(|I| - 1)$, some member of some $\mathcal{A}_i$ with $i \in I$ is disjoint from S . Then there are pairwise disjoint sets $A_1 \in \mathcal{A}_1, \dots, A_q \in \mathcal{A}_q$.*

In Haxell's formulation, the families form a bipartite hypergraph in which each edge consists of one index i together with a member of $\mathcal{A}_i$, so edges have at most $r = h + 1$ vertices, and the bound $(2r - 3)(|I| - 1)$ on transversals is the bound above.

We also need a Bernstein-type inequality for functions of independent bits. It follows by applying Freedman's martingale inequality [8] to the exposure martingale; see also [18].

Lemma A.2 (Bernstein inequality for bounded differences). *Let $\xi_1, \dots, \xi_k$ be independent $\{0, 1\}$ -valued random variables with $\mathbb{P}(\xi_i = 1) = p_i$, and let $Z = \phi(\xi_1, \dots, \xi_k)$, where changing ξ_i alone changes ϕ by at most $c_i \leq c$. Put $\sigma^2 = \sum_i p_i(1 - p_i)c_i^2$. Then for all $x > 0$,*

$$\mathbb{P}(Z \leq \mathbb{E}Z - x) \leq \exp\left(-\frac{x^2}{2(\sigma^2 + cx/3)}\right).$$

Proof. Let $Z_i = \mathbb{E}[Z \mid \xi_1, \dots, \xi_i]$. Given $\xi_1, \dots, \xi_{i-1}$, let $g(y) = \mathbb{E}[Z \mid \xi_1, \dots, \xi_{i-1}, \xi_i = y]$. By independence $g(1) - g(0)$ is an average of differences of ϕ , so $|g(1) - g(0)| \leq c_i$. Hence the martingale difference $Z_i - Z_{i-1} = g(\xi_i) - \mathbb{E}g(\xi_i)$ is bounded by c in absolute value and has conditional variance $p_i(1 - p_i)(g(1) - g(0))^2 \leq p_i(1 - p_i)c_i^2$. Freedman's inequality [8], applied to the martingale $-Z_i$, now gives the bound. $\square$

Setup

Let H, s, t, ρ and V be as in [Lemma 4.3](#). Write $L = \log N$ and $\alpha = \varepsilon/L^2$. Since $N > s \geq C_* L^{18}$, the order N is as large as we like if C_* is large.

For a graph H' on $V(H)$, a set U , a set F of edges and an integer $k \geq 0$, the *ball* $\mathcal{B}_{H'-F}^k(U)$ is the set of vertices of V that can be reached from U by a path in $H' - F$ through V of length at most k . Paths of length 0 are allowed, so $U \cap V \subseteq \mathcal{B}_{H'-F}^k(U)$. We use the parameters

$$k = \left\lceil \frac{100L}{\alpha} \right\rceil, \quad p = \frac{\rho}{10k}, \quad \theta = \frac{C_1 L}{\alpha p}, \quad \zeta = \frac{\alpha}{6\theta}, \quad h = \lceil 4kL \rceil, \quad \bar{s} = \frac{40ht}{\rho}, \quad K = \left\lceil \frac{\bar{s}}{\zeta} \right\rceil,$$

where C_1 is a large absolute constant. Here k is the radius of the balls, p the density of one batch of V , θ the expansion of well-expanding sets, $\zeta|U|$ the number of deleted edges that one colour class tolerates, h the length of the paths, $\bar{s}|U|$ the number of deleted edges needed for Haxell's theorem, and K the number of colours. Then $k = O(L^3)$, $\theta = O(L^6/\rho)$, $\zeta^{-1} = O(L^8/\rho)$, $h = O(L^4)$ and $K = O(tL^{12}\rho^{-2})$. So $2K(\theta + 1) = O(tL^{18}\rho^{-3}) \leq s$ if C_* is large.

From balls to paths

The first claim is [4, Proposition 8].

Claim A.3. Let H' be a graph on $V(H)$ and $u \geq 1$ with $|V| \geq 4u - 2$, and suppose $|\mathcal{B}_{H'}^k(U)| > |V|/2$ for every set U of u vertices. Let $\{x_1, y_1\}, \dots, \{x_{2u-1}, y_{2u-1}\}$ be pairs, all of whose $4u - 2$ vertices are distinct. Then some x_i and y_i are joined by a path in H' through V of length at most $4kL$.

Proof. Call x *good* if $|\mathcal{B}_{H'}^{2kL}(\{x\})| > |V|/2$. If both x_i and y_i are good, their balls meet in a vertex $z \in V$, and joining the two paths at z gives a walk through V of length at most $4kL$. This walk contains the required path. So it suffices to show that fewer than u vertices are bad.

Suppose S_0 is a set of u bad vertices. We find sets $S_0 \supseteq S_1 \supseteq \dots$ with $|S_{i+1}| \leq \lceil |S_i|/2 \rceil$ and $|\mathcal{B}_{H'}^{(i+1)k}(S_i)| > |V|/2$, starting from the hypothesis for S_0 . Given S_i with $|S_i| \geq 2$, split it into two halves. The ball of S_i is the union of the balls of the halves, so one half S_{i+1} has $|\mathcal{B}_{H'}^{(i+1)k}(S_{i+1})| > |V|/4 \geq u - \frac{1}{2}$. Choose u vertices in this ball. They lie in V , so their own ball of radius k has more than $|V|/2$ vertices. Concatenating paths shows that this ball lies in $\mathcal{B}_{H'}^{(i+2)k}(S_{i+1})$. After $j \leq \lceil \log u \rceil \leq L + 1$ steps we reach a single vertex $x \in S_0$ with $|\mathcal{B}_{H'}^{(j+1)k}(\{x\})| > |V|/2$, and $(j+1)k \leq 2kL$. So x is good, which is a contradiction. $\square$

Claim A.4. Suppose that $|V| \geq \rho N/2$ and that $|\mathcal{B}_{H-F}^k(U)| > |V|/2$ for all nonempty $U \subseteq V(H)$ and all $F \subseteq E(H)$ with $|F| \leq \bar{s}|U|$. Then H links through V with multiplicity t .

Proof. Let $\pi_1, \dots, \pi_q$ be a multiset of pairs in which every vertex lies in at most t pairs. Let $\mathcal{A}_i$ be the family of edge sets of paths through V of length at most h joining the pair π_i . We check the condition of [Theorem A.1](#). Let $I \subseteq [q]$ be nonempty and let S be a set of edges with $|S| \leq (2h-1)(|I|-1)$. Choose a maximal $I' \subseteq I$ whose pairs are pairwise disjoint, and let $q' = |I'|$. Every pair of I meets one of the $2q'$ vertices covered by I' , and each of these lies in at most t pairs, so $|I| \leq 2tq'$. Let $u = \lceil \rho q'/10 \rceil$. Then $2u - 1 \leq q'$, and $4u - 2 \leq \rho N/5 + 2 \leq |V|$, because $q' \leq N/2$. Moreover

$$|S| < 2h|I| \leq 4htq' \leq \frac{40ht}{\rho}u = \bar{s}u.$$

So every set U of u vertices has $|\mathcal{B}_{H-S}^k(U)| > |V|/2$. Apply [Claim A.3](#) to $H' = H - S$ and $2u - 1$ of the pairs in I' . It gives a path of length at most $4kL \leq h$ in $H - S$ through V that joins some pair π_i with $i \in I$. Its edge set lies in $\mathcal{A}_i$ and avoids S . By [Theorem A.1](#) we can choose pairwise disjoint members of the $\mathcal{A}_i$, that is, edge-disjoint paths. $\square$

Large balls

In [Claims A.5–A.9](#), H' is an s' -expander on $V(H)$ with $s' \geq \theta + 1$. The next claim replaces the case analysis of [4, Proposition 13].

Claim A.5 (A growth certificate). Let $1 \leq |W| \leq 2N/3$, $F \subseteq E(H')$ with $|F| \leq s'|W|/4$, and let b be an integer with $1 \leq b \leq s'$. Let Φ be a maximal set of edges of $H' - F$ between W and $V(H) \setminus W$ in which every vertex has degree at most b . Then $|\Phi| \geq \alpha b|W|/4$.

Proof. Suppose not. Then the sets $W_b \subseteq W$ and $X_b \subseteq V(H) \setminus W$ of vertices of Φ -degree exactly b each have fewer than $\alpha|W|/4$ elements. Let $U = W \setminus W_b$. By maximality, every edge of $H' - F - \Phi$ from U to $V(H) \setminus W$ ends in X_b . So $N_{H'-F-\Phi}(U) \subseteq W_b \cup X_b$, a set of fewer than $\alpha|W|/2$ vertices. But $|U| \geq 3|W|/4$ and $|F \cup \Phi| \leq s'|W|/4 + \alpha b|W|/4 \leq s'|W|/2 \leq s'|U|$. So expansion gives $|N_{H'-F-\Phi}(U)| \geq \alpha|U| \geq 3\alpha|W|/4$, which is a contradiction. $\square$

Claim A.6 (One random batch). In the setting of [Claim A.5](#) with $b = \lceil 1/p \rceil$, let $S \subseteq W$ contain each vertex independently with probability p . Then, with probability at least $1 - \exp(-\alpha p|W|/300)$, at least $\alpha|W|/16$ vertices outside W have an $(H' - F)$ -neighbour in S .

Proof. Let Z be the number of vertices outside W with a Φ -neighbour in S . A vertex of Φ -degree $i \geq 1$ is counted with probability $1 - (1 - p)^i \geq pi/2$, since $pi \leq pb \leq 1 + p \leq 3/2$. So $\mathbb{E}Z \geq p|\Phi|/2 \geq \alpha|W|/8$. Changing whether $w \in W$ lies in S changes Z by at most $\deg_\Phi(w) \leq b$, and $\sigma^2 = \sum_w p(1 - p) \deg_\Phi(w)^2 \leq pb|\Phi| \leq 2b\mathbb{E}Z$. By [Lemma A.2](#) with $x = \mathbb{E}Z/2$ and $b \leq 2/p$,

$$\mathbb{P}(Z < \tfrac{1}{2}\mathbb{E}Z) \leq \exp\left(-\frac{3\mathbb{E}Z}{52b}\right) \leq \exp\left(-\frac{\alpha p|W|}{300}\right). \quad \square$$

Call a set A *well-expanding* if $|N_{H'}(A)| \geq \theta|A|$.

Claim A.7 (Sprinkling). Let A be well-expanding with $a = |A| \geq 1$, and let $F \subseteq E(H')$ with $|F| \leq a$. Then $\mathbb{P}(|\mathcal{B}_{H'-F}^k(A)| \leq 0.56\rho N) \leq N^{-10a}$.

Proof. Realise V as $V_1 \cup \dots \cup V_k$ with independent random sets: V_i is p -random for $i < k$, and V_k is q -random with $(1 - p)^{k-1}(1 - q) = 1 - \rho$. As $(k - 1)p \leq \rho/10$, we have $q \geq 0.9\rho$. For $1 \leq i \leq k$ let $\mathcal{R}_i$ be the set of vertices reachable from A by a path in $H' - F$ of length at most i through $V_1 \cup \dots \cup V_{i-1}$. Then $\mathcal{R}_1 = A \cup N_{H'-F}(A)$ has at least θa vertices, since deleting a edges removes at most a neighbours. If a vertex outside $\mathcal{R}_i$ has an $(H' - F)$ -neighbour in $\mathcal{R}_i \cap V_i$, then it lies in $\mathcal{R}_{i+1}$.

Condition on $V_1, \dots, V_{i-1}$ and suppose $|\mathcal{R}_i| \leq 2N/3$. [Claim A.6](#) applies with $W = \mathcal{R}_i$: we have $|F| \leq a \leq s'|\mathcal{R}_i|/4$ and $b = \lceil 1/p \rceil \leq \theta + 1 \leq s'$. So $|\mathcal{R}_{i+1}| \geq (1 + \alpha/16)|\mathcal{R}_i|$, except with probability at most $\exp(-\alpha p\theta a/300) = \exp(-C_1 L a/300) \leq N^{-12a}$. Since $(1 + \alpha/16)^{k-1} > N$, with probability at least $1 - kN^{-12a}$ we reach $|\mathcal{R}_k| > 2N/3$. Every vertex of $\mathcal{R}_k \cap V_k$ lies in $\mathcal{B}_{H'-F}^k(A)$. Given $|\mathcal{R}_k| > 2N/3$, the size of $\mathcal{R}_k \cap V_k$ is binomial with mean at least $0.6\rho N$. So by [Lemma 2.3](#) it is at most $0.56\rho N$ with probability at most $e^{-\rho N/750} \leq e^{-\rho\theta a/750} \leq N^{-11a}$. Here we used $N \geq \theta a$ and $\rho\theta \geq 10^4 C_1 L$. $\square$

Claim A.8 (Well-expanding subsets). Every U with $1 \leq |U| \leq 2N/3$ contains a well-expanding set A with $|A| \geq \alpha|U|/(3\theta) = 2\zeta|U|$.

Proof. This is [\[4, Proposition 18\]](#). Let $A \subseteq U$ be a maximal well-expanding set ($\emptyset$ is well-expanding). If $A = U$ we are done. Otherwise, by maximality every $v \in U \setminus A$ has fewer than $\theta + 1$ neighbours outside $A \cup N_{H'}(A)$; adding v to A would otherwise create a larger well-expanding set. Delete these fewer than $(\theta + 1)|U| \leq s'|U|$ edges. Afterwards the external neighbourhood of U lies in $N_{H'}(A)$, so expansion gives $\alpha|U| \leq |N_{H'}(A)|$. In particular $A \neq \emptyset$. On the other hand, $A \cup \{v\}$ is not well-expanding, so $|N_{H'}(A)| < \theta(|A| + 1) + 1 \leq 3\theta|A|$. $\square$

Claim A.9. With probability at least $1 - 2N^{-6}$, $|\mathcal{B}_{H'-F}^k(U)| > 0.56\rho N$ for every nonempty U and every $F \subseteq E(H')$ with $|F| \leq \zeta|U|$.

Proof. Take a union bound of [Claim A.7](#) over all pairs (A, F) with A well-expanding, $|A| = a$ and $|F| \leq a$. There are at most N^{3a+1} such pairs for each a , so the failure probability is at most $\sum_a N^{1-7a} \leq 2N^{-6}$. If $|U| \leq 2N/3$, [Claim A.8](#) gives a well-expanding $A \subseteq U$ with $|A| \geq 2\zeta|U| \geq |F|$, and $\mathcal{B}_{H'-F}^k(U) \supseteq \mathcal{B}_{H'-F}^k(A)$. If $|U| > 2N/3$, apply this to a subset $U' \subseteq U$ of size $\lceil N/2 \rceil$, for which $|F| \leq \zeta N \leq 2\zeta|U'|$. $\square$

Amplification

Colour the edges of H uniformly at random with K colours, independently of V , and let $H_1, \dots, H_K$ be the colour classes. Consider the events

- (E1) $\rho N/2 \leq |V| \leq 1.1\rho N$;
- (E2) every H_i is an $(s/2K)$ -expander, where $s/2K \geq \theta + 1$;
- (E3) the conclusion of [Claim A.9](#) holds for every $H' = H_i$.

Note that $\rho N \geq (C_* L^{18} N^2)^{1/3}$, because $\rho^3 \geq C_* L^{18}/s$ and $s < N$. By [Lemma 2.3](#), [Lemma 4.2](#) with $p = 1/K$, and [Claim A.9](#), all three events hold with probability at least $1 - 2e^{-\rho N/300} - KN^{-10} - 2KN^{-6} \geq 1 - N^{-3}$, since $K < s < N$. Assume they hold. Let U be nonempty and $|F| \leq \bar{s}|U|$. Some colour i has $|F \cap E(H_i)| \leq \bar{s}|U|/K \leq \zeta|U|$, so

$$|\mathcal{B}_{H-F}^k(U)| \geq |\mathcal{B}_{H_i-F}^k(U)| > 0.56\rho N \geq |V|/2.$$

By [Claim A.4](#), H links through V with multiplicity t . The colouring was only an auxiliary device. Linking is a property of V alone, and it fails with probability at most N^{-3} . This proves [Lemma 4.3](#). $\square$

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